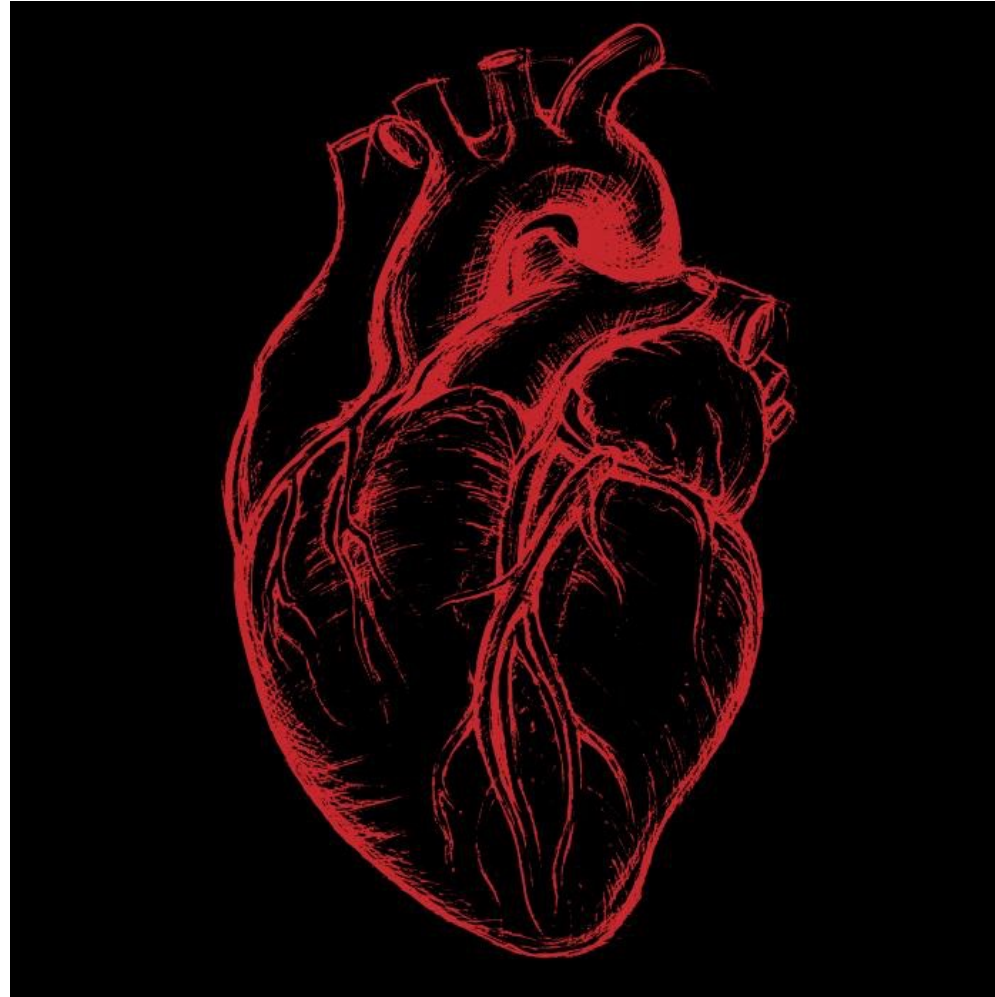


Circulatory Systems



Circulatory Systems

Key Concepts:

- **Types of Circulatory Systems**
- **The Composition of Blood**
- **The Vertebrate Heart and Its Function**
- **Blood Vessels**
- **Relationship Among Blood Pressure, Blood Flow, and Resistance**
- **Impact on Public Health**

Types of Circulatory Systems

Primary function of a circulatory system is to **transport**

- **Necessary materials to the cells** of an animal's body
- **Waste products away** from the cells to be released into the environment

Two basic **types**:

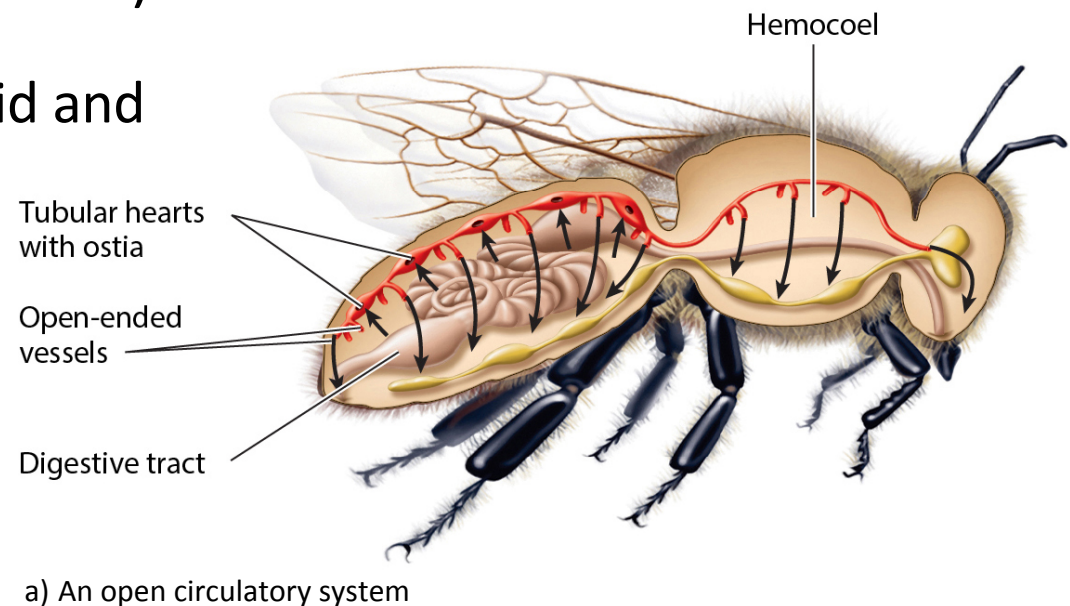
- **Open** circulatory system
- **Closed** circulatory system

Open Circulatory Systems

Arthropods and some mollusks

Basic components:

- **Hemolymph** (instead of blood)
 - Mixture of circulatory fluid and interstitial fluid
- **One or more hearts**
- **Vessels**
 - **open** into animal's **body cavity** (hemocoel)



Pros and Cons of Open Circulation

Nutrients and waste exchange by diffusion between hemolymph and body cells

Metabolically inexpensive

As an animal's activity increases, circulation becomes more efficient

Important Limitation: hemolymph cannot be selectively delivered to different tissues

Closed Circulatory Systems

Blood (circulatory fluid) remains **within vessels for distribution**

- **Arteries** carry blood **away from heart**
- **Veins** bring it **back**,
- **capillaries** link arteries and veins

Blood and interstitial fluid are **physically separated**

- **Differ in components** and chemical composition

Found in **earthworms, cephalopods, and all vertebrates**

- **Some animals** have a closed circulatory system composed of a **single circuit**
- **Others** have **double circulation**

Typical Features of Closed Circulation

One or more contractile, **muscular hearts**

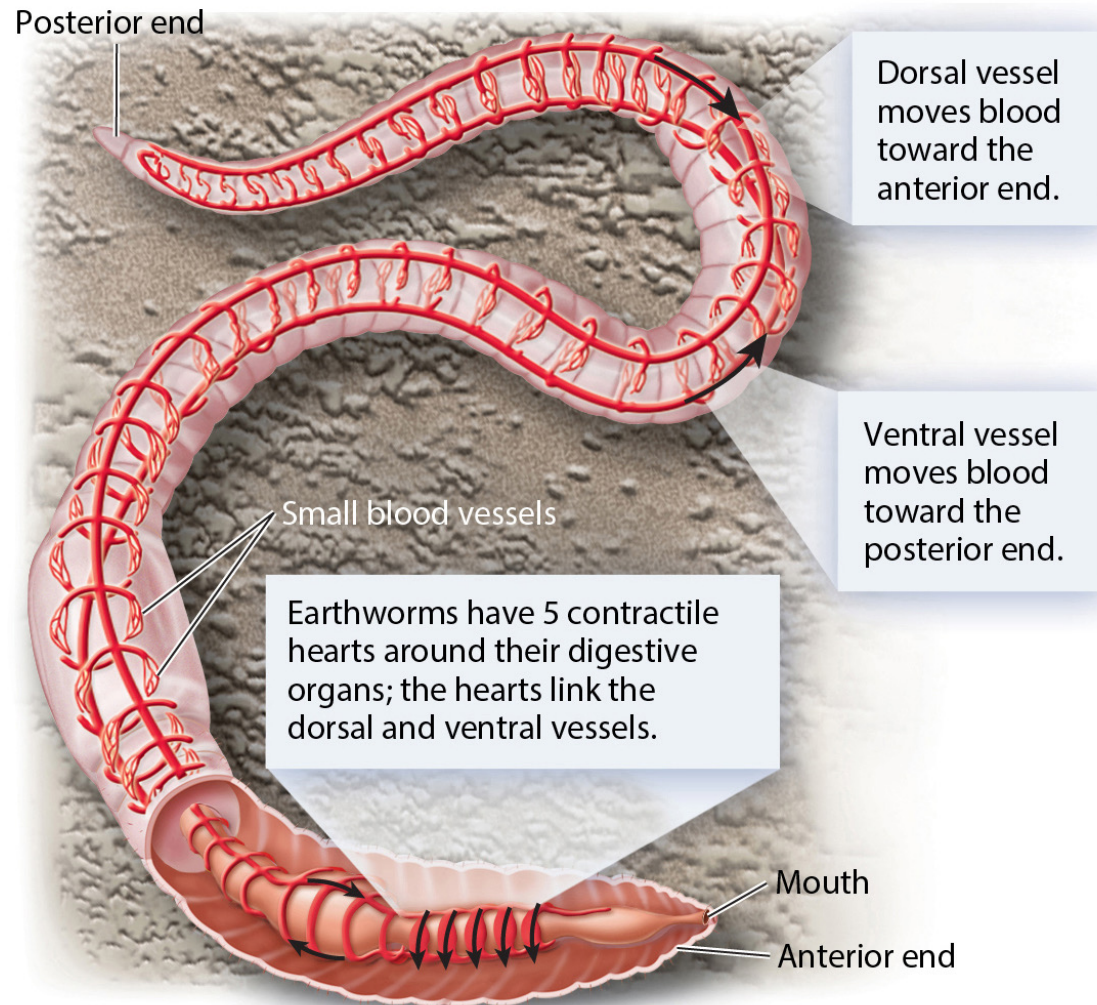
Solutes exchanged with environment and body cells

Adjusted to match the animal's **metabolic demands**

Capacity to heal themselves when wounded (clots)

A Closed Circulatory System

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b) The closed **circulatory system** of earthworms.

Advantages of Closed Circulation

Animal can grow larger with more efficient supply

- Nearly all **body cells are within one or two cell widths of a blood vessel**
- System grows in size as an animal grows

Blood flow can be selectively controlled

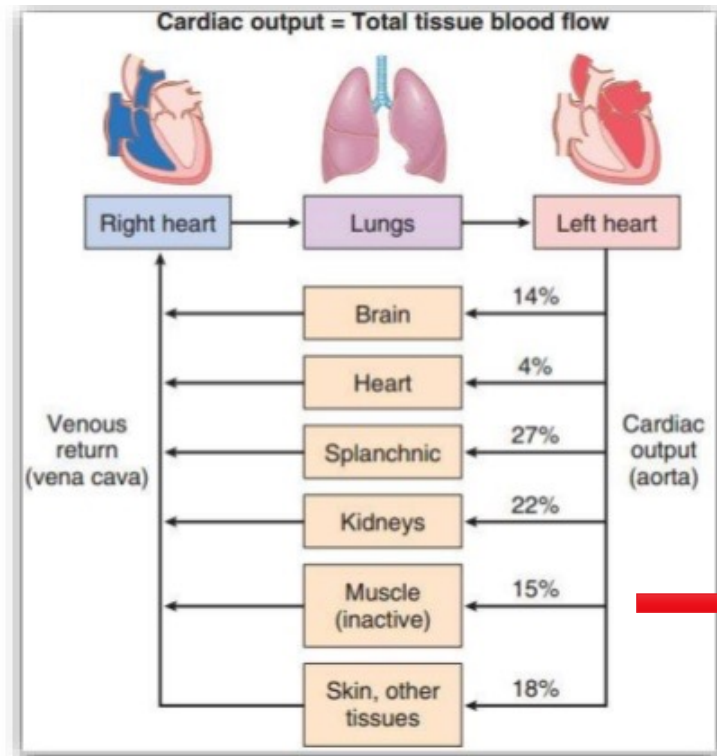
- Vasodilation/vasoconstriction select which blood vessels are carrying blood at any given time
- **Different body parts can be supplied with the precise amount of blood needed at any given moment**

Blood distribution is increased in exercising muscles

With **maximal exercise** in humans **blood flow** to :

- **skeletal muscle is dramatically increased**

Distribution of cardiac output at rest

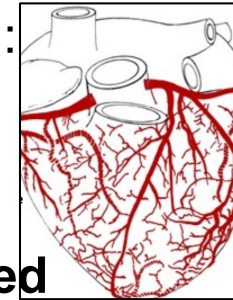


Increases to 80–85% during maximal exercise

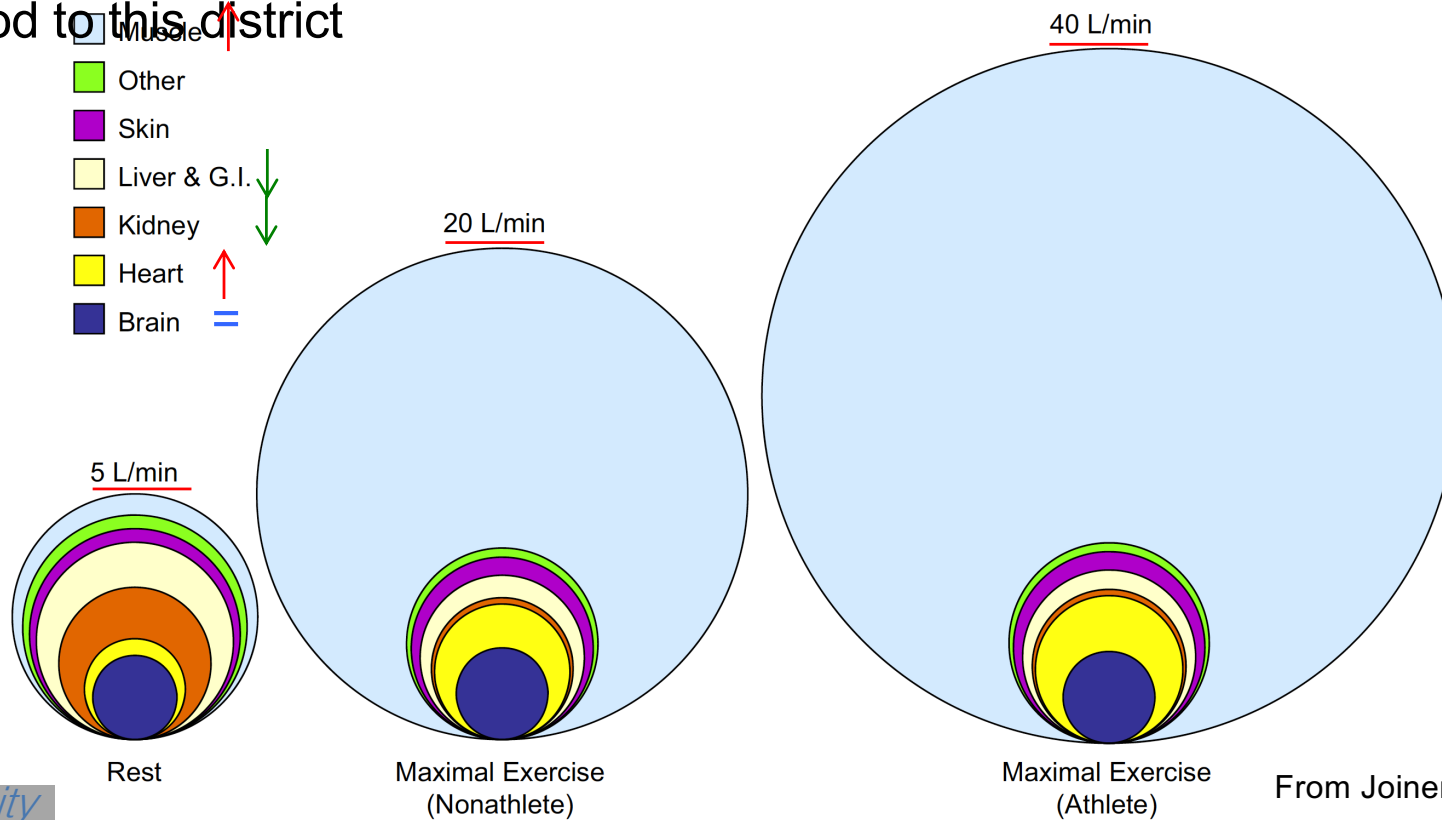
Blood tissue distribution is altered during exercise

With **maximal exercise** in humans **blood flow** to :

- **Skeletal muscle** is **dramatically increased**
- **Coronary** arteries (heart) is **increased**
- **Brain** is **constant**
- Internal organs such **kidney, liver, GI** is **reduced**
- **Blood flow to the skin** is **relatively low** unless in conditions of **high temperature** when cutaneous dilation induce a significant redistribution of blood to this district



Coronary arteries

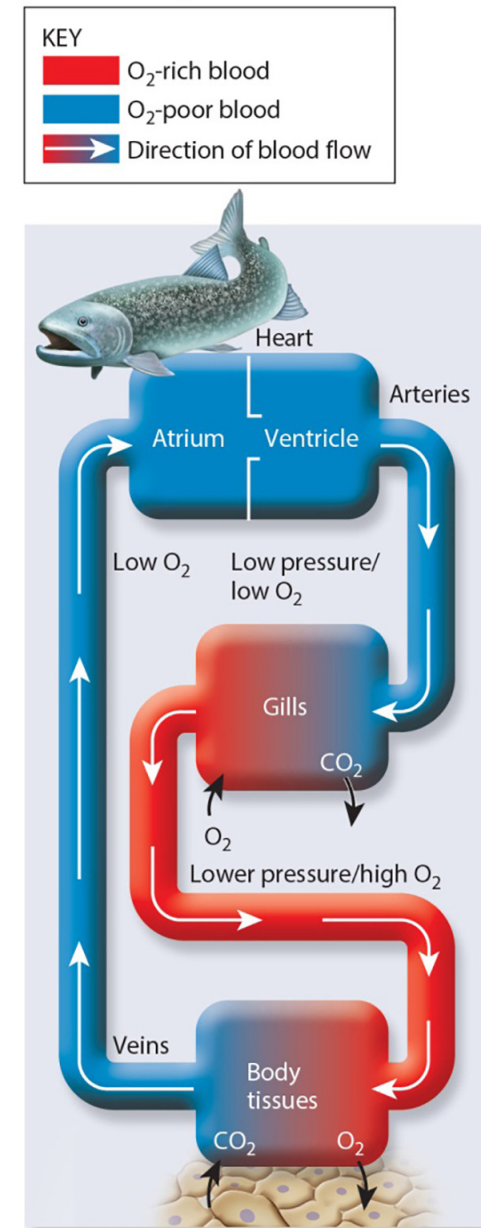


From Joiner and Casey 2015

Single Circulation 1

- Found in **fishes**
- **Single atrium collects blood** from tissues
- **Single ventricle pumps blood** out of the heart
- **Arteries carry blood away from the heart** to the **gills**
- **Blood picks up oxygen, drops off carbon dioxide** in gills
- Freshly oxygenated blood continues on through arteries to other body tissues
- Oxygen and nutrients delivered to tissues, carbon dioxide diffuses into blood
- **Partially deoxygenated blood returns to the heart** via **veins**

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(a) Single circulation in fishes

Single Circulation Limitation: low pressure

Respiratory surfaces function **best** when the blood flowing through them is maintained **at a low pressure**

Single **fish heart does not generate high pressure** when pumping blood to gills

→ This means that **blood pressure leaving the gills will also be low**, which **limits rate** at which **oxygenated blood can be delivered to body**

Double Circulation

- Found in **crocodiles, birds and mammals**
- **Oxygenated and deoxygenated blood separates into two distinct circuits**
 - **Two atria**
 - **Two ventricles**

Pulmonary circulation

- Receives deoxygenated blood from right ventricle
- Delivers blood to lungs

Systemic circulation – to the body

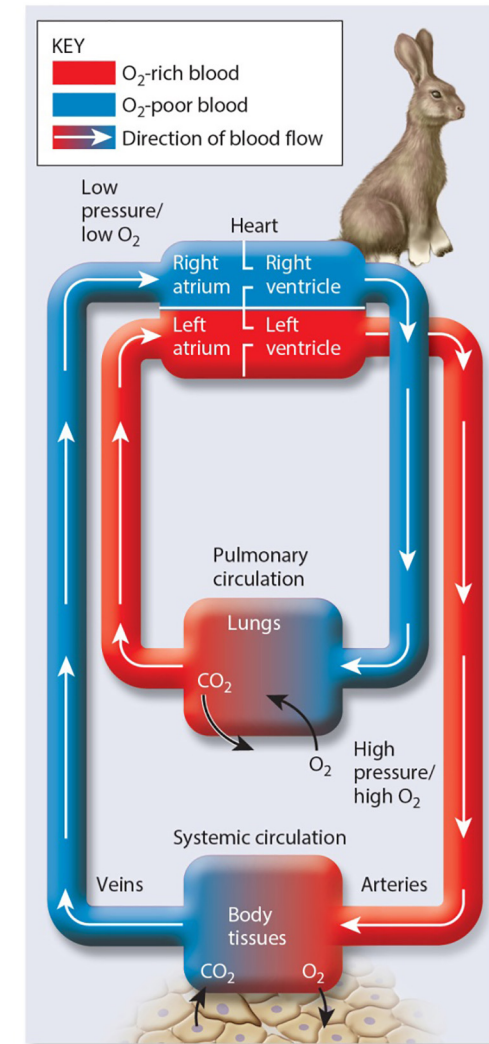
- Receives oxygenated blood from left ventricle
- Delivers blood to body

Double Circulation of Mammals

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Major advantage of double circulation is **two different blood pressures** in two different systems:

- **Right ventricle** can pump blood through **pulmonary circuit** with **low pressure**
- **Left ventricle** generates **higher pressure** into **systemic circulation**



(b) Double circulation in crocodiles, birds, and mammals

[Access the text alternative for slide images.](#)

The Composition of Blood

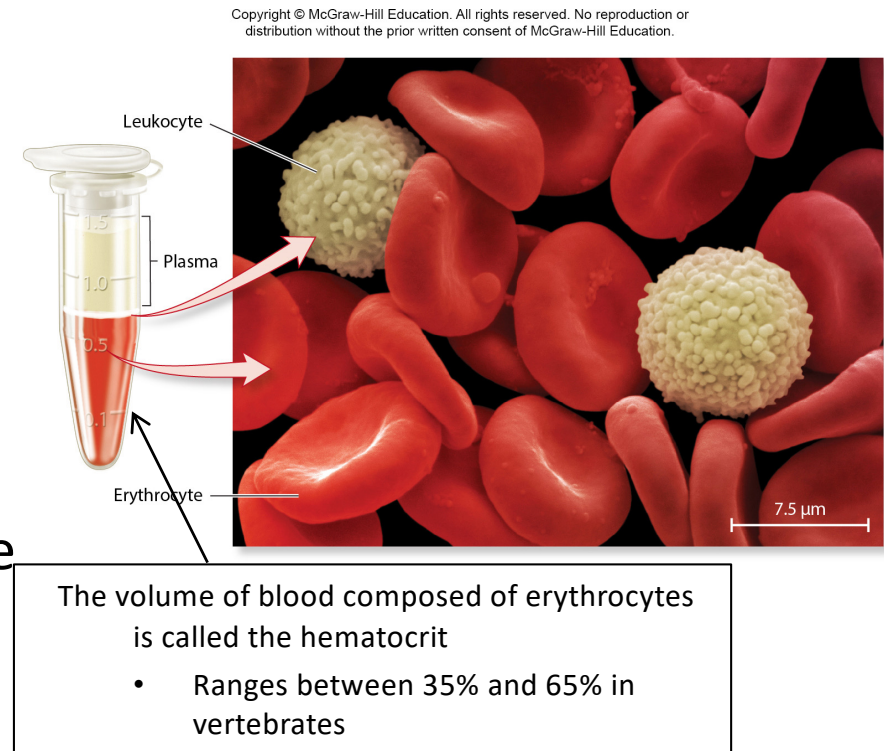
Blood is the **transport medium** of animals with closed circulatory systems.

-Plasma

- **Water and dissolved substances** (including nutrients, oxygen, waste products, hormones, proteins)

-Cells

- **Erythrocytes** (red blood cells) serve the critical function of **transporting oxygen**
- **Leukocytes** (white blood cells) are crucial components of the **immune system**
- **Platelets** are cell fragments that function in formation of blood clots

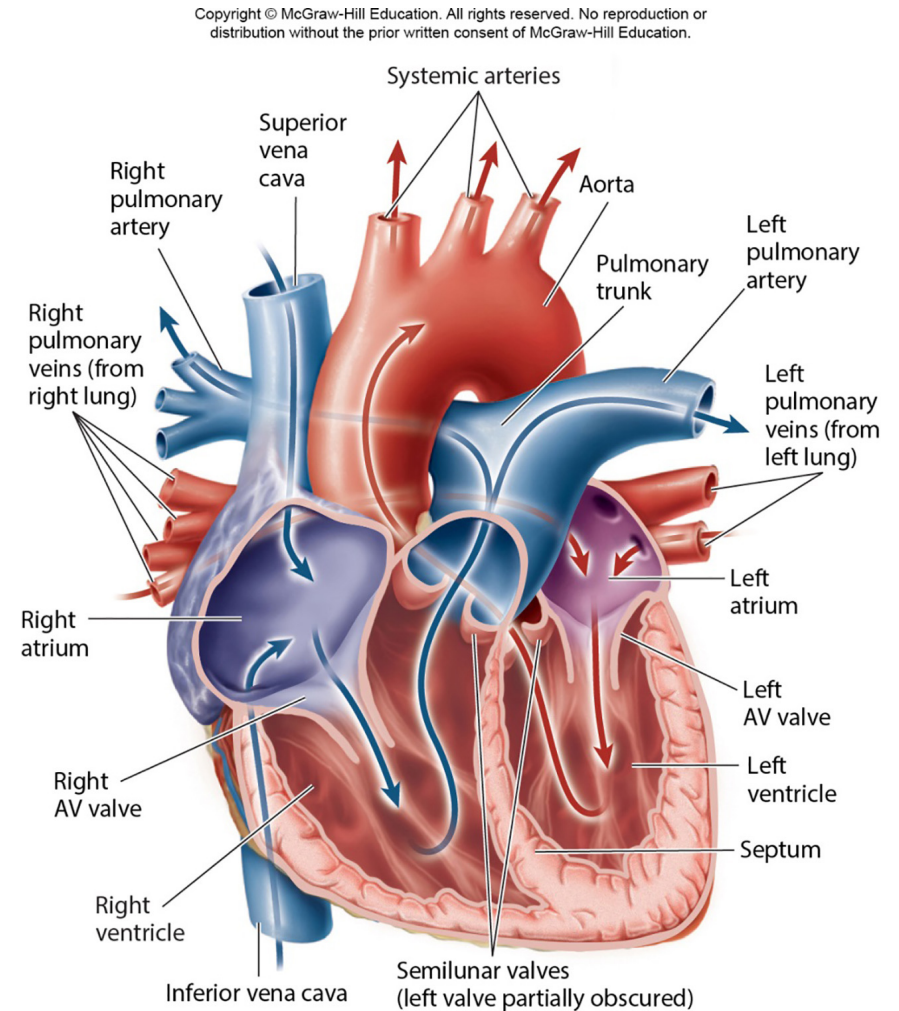


The Vertebrate Heart

Two sides of the heart are separated by a wall called the **septum**

Double circulation with **four** chambers

- **Right atrium**
- **Right ventricle**
- **Left atrium**
- **Left ventricle**



Flow of Blood Through Heart ¹

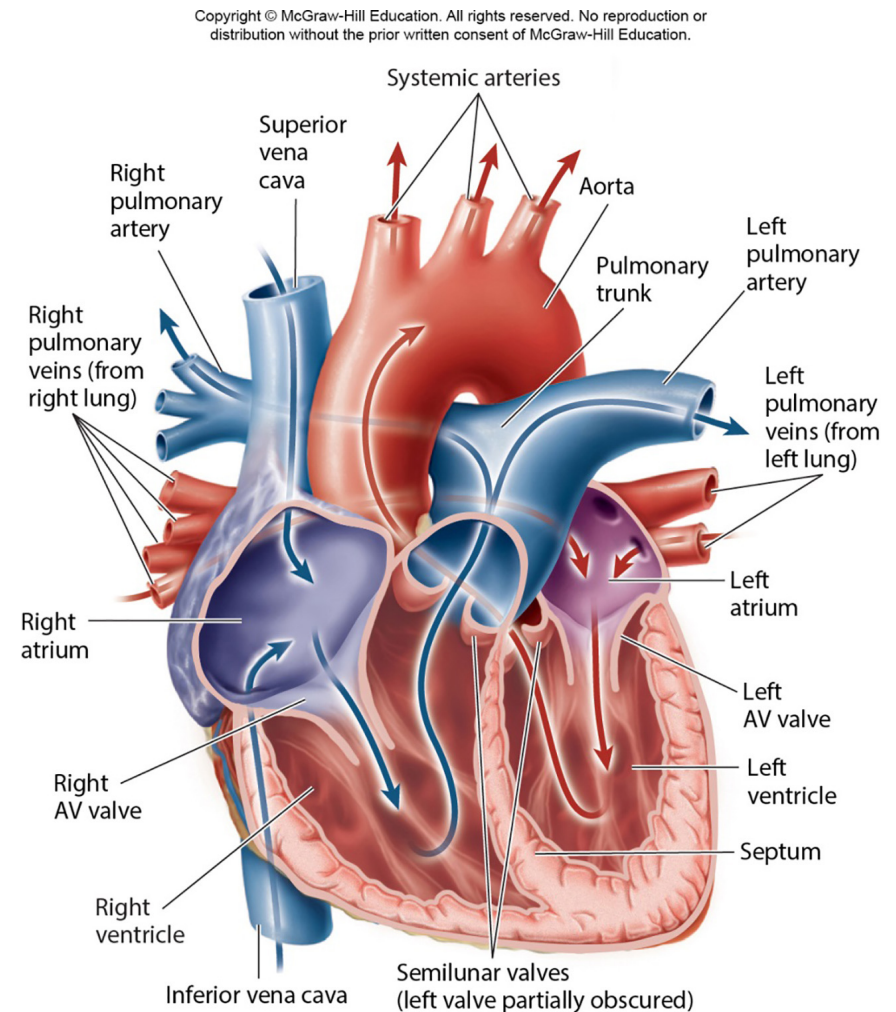
Blood enters from **systemic** (inferior or superior vena cava) or **pulmonary veins** into **atrium**

Left ventricle ejects blood into the aorta - branches to arteries that distribute the blood

Right ventricle ejects blood into the pulmonary trunk – divides into pulmonary arteries

One-way valves:

- **atrioventricular (AV) valves** into ventricles
- **semilunar valves** into **systemic** (aorta) or **pulmonary arteries**



Myogenic Hearts

Electrically excitable, generates its own action potentials

- **Beats on its own**
- In contrast, neurogenic hearts of arthropods require regular electrical impulses from the nervous system

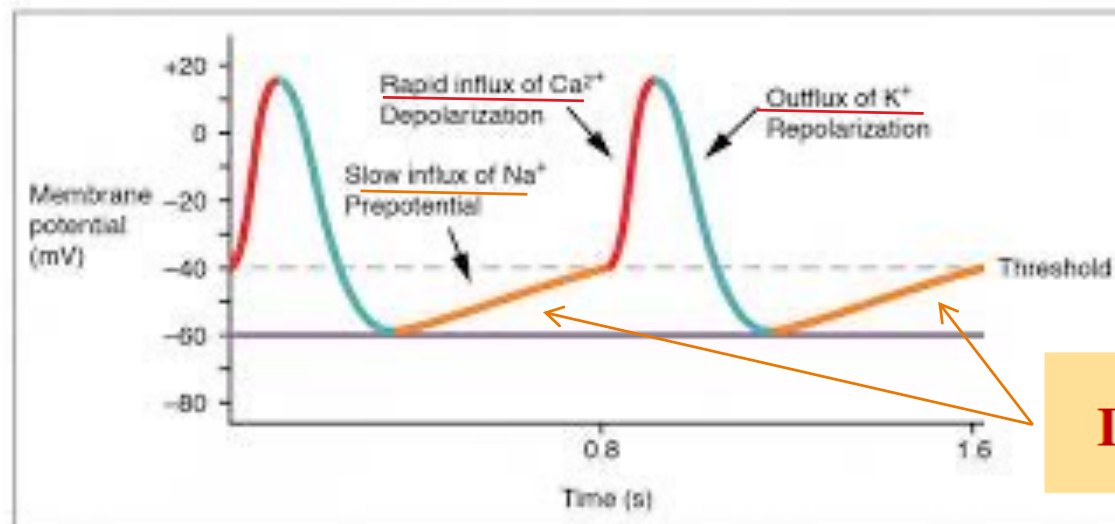
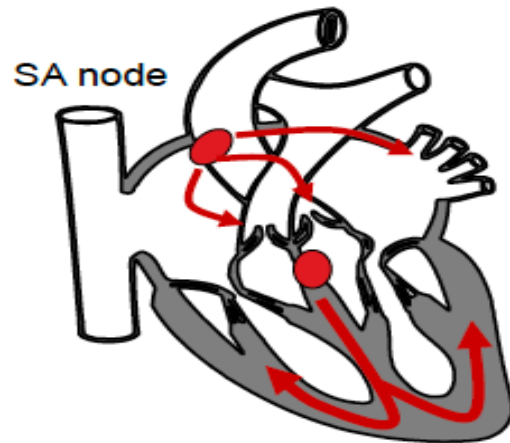
Nervous input (autonomic) can increase or decrease rate

Sinoatrial node (SA node) functions as pacemaker

- **Collection of modified cardiac cells that spontaneously and rhythmically generate action potentials**

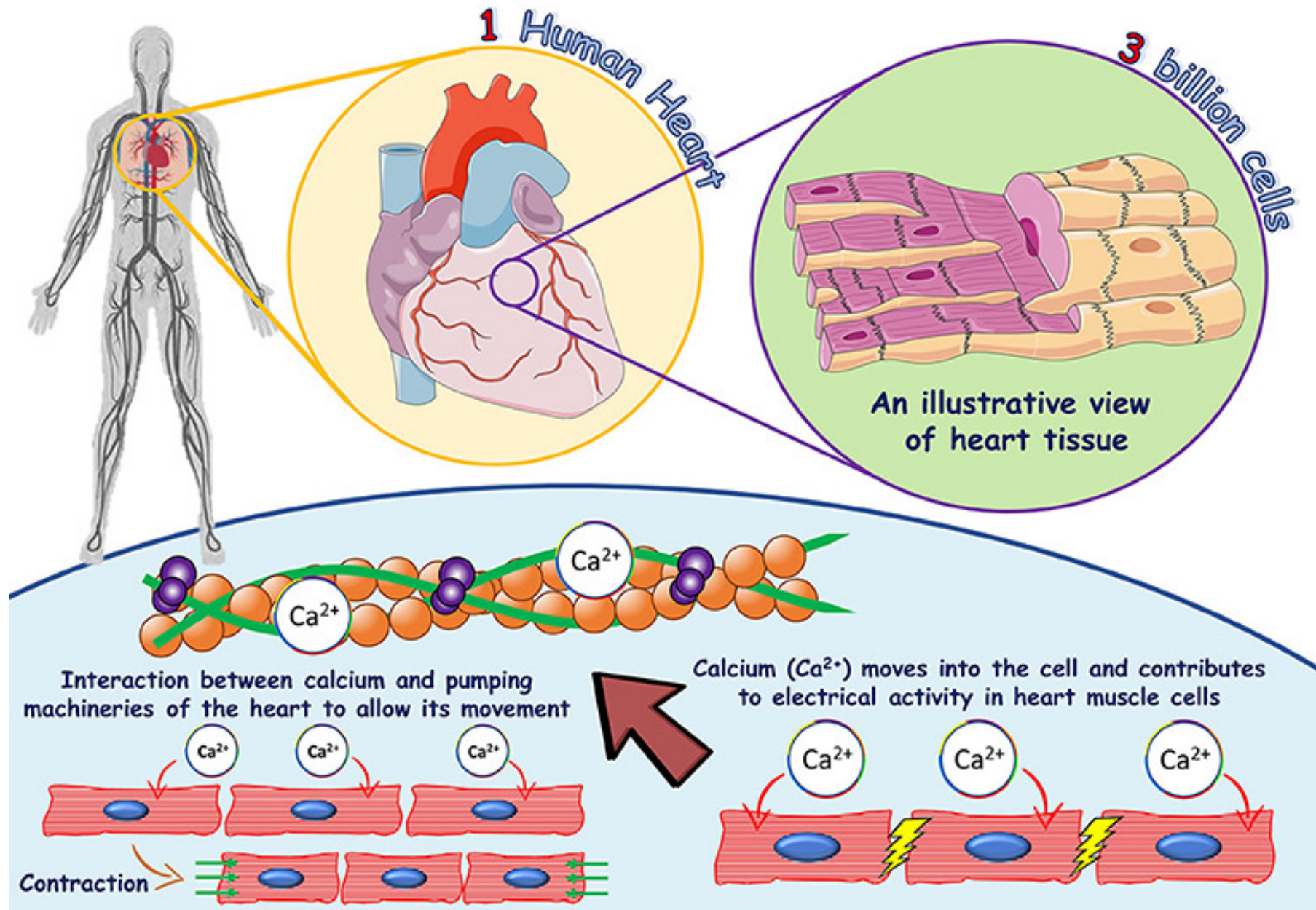
Action potential in the Sinoatrial (SA) node

In the cells of the SA node **absence of stable resting potential** during diastole (→ **diastolic depolarization**)

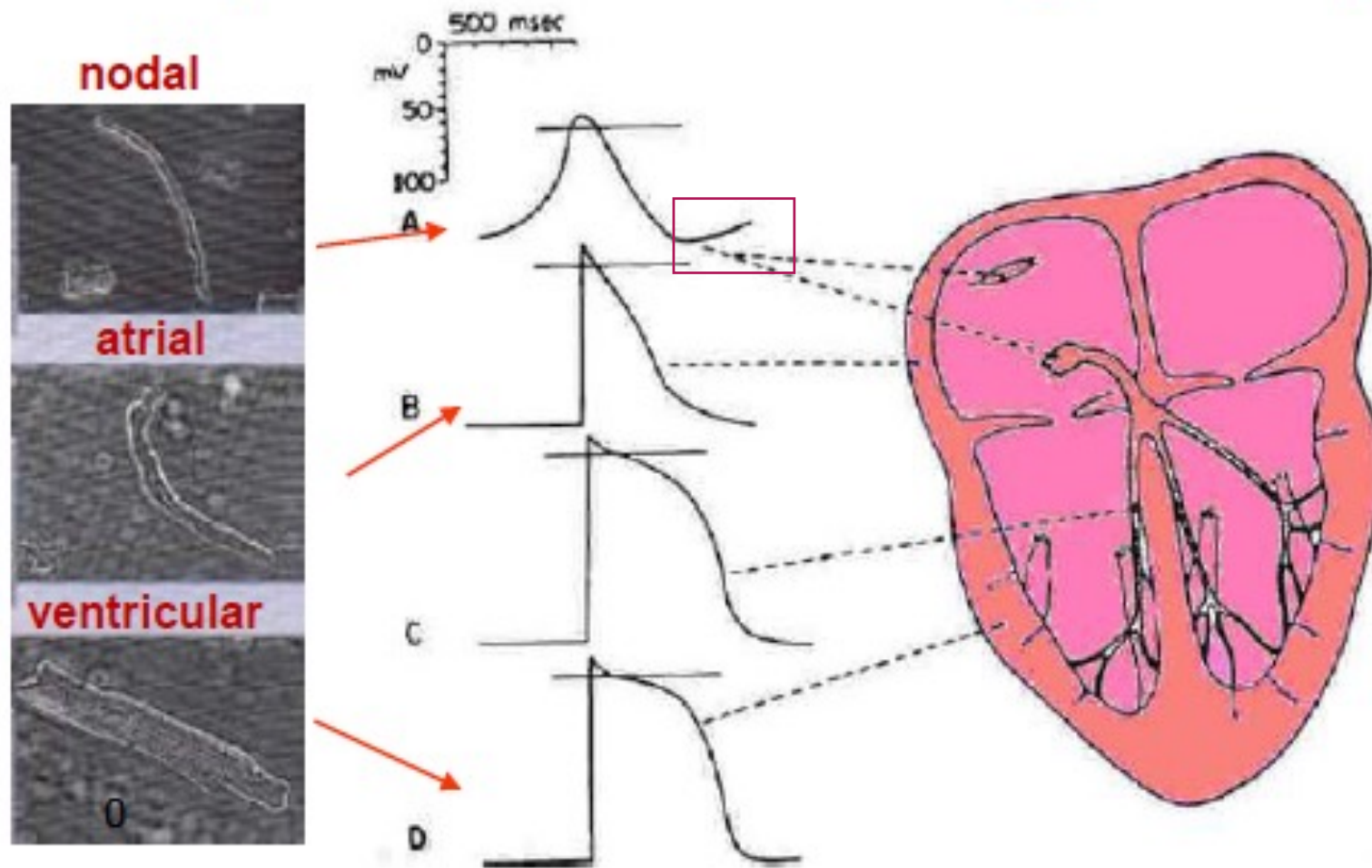


Diastolic depolarization

Calcium and cardiac contraction



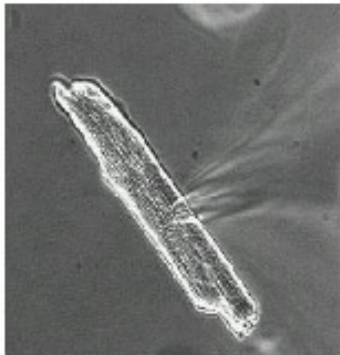
Action potential heterogeneity



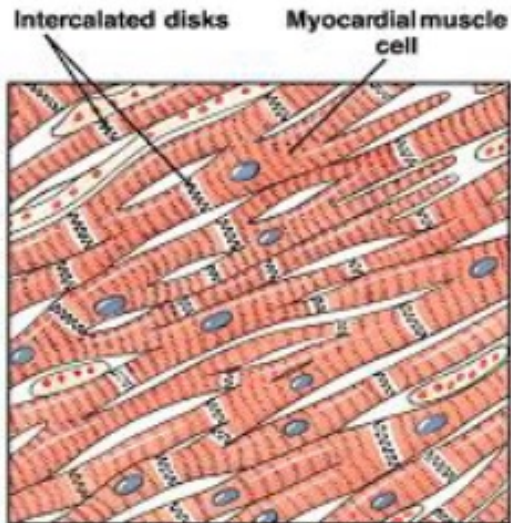
Propagation of electrical waves

Action potentials spread because cardiac cells are electrically coupled by gap junctions (syncytium)

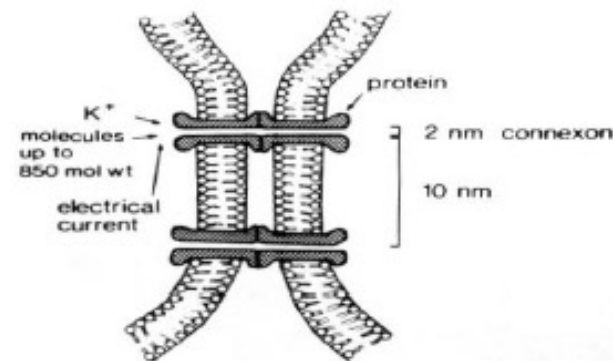
Cardiac cell



- cylindrical shape
- striated
- 50-100 μm length
- 10-25 μm diameter

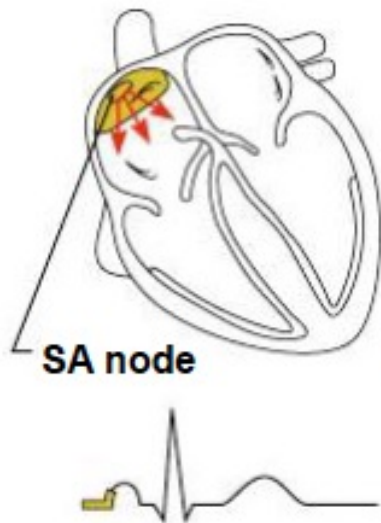


Gap junctions

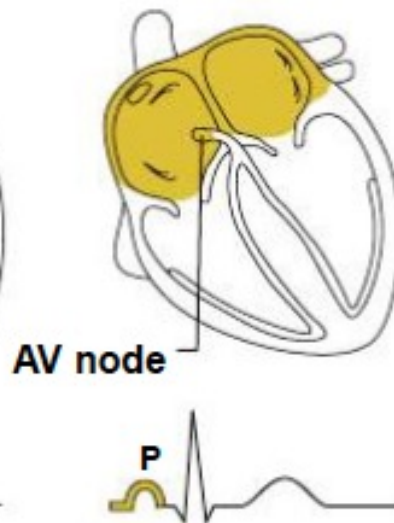


Propagation of electrical waves

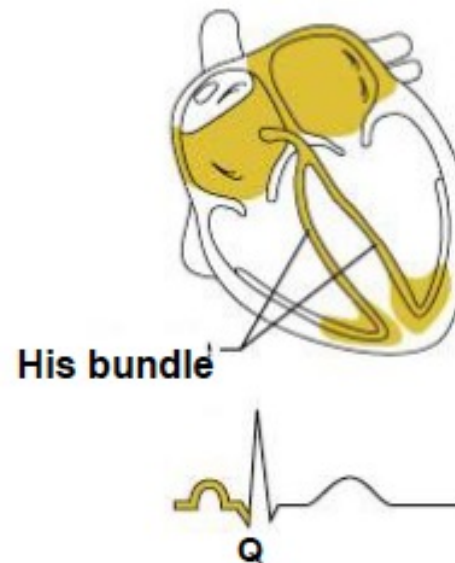
Impulse generation in the SA node: atrial excitation begins



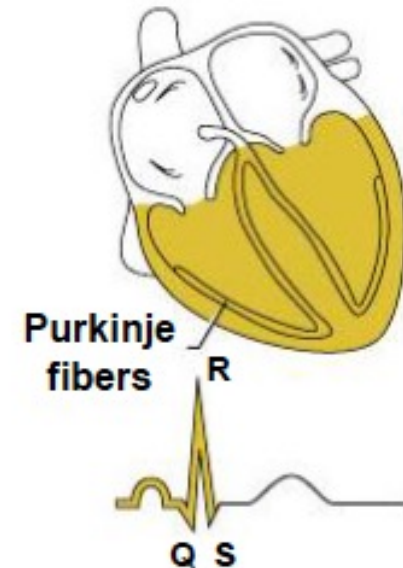
The wave of excitation spreads across the atria and reaches the AV node, where the signal is delayed



Excitation is then conducted into the bundle of His: ventricular excitation begins



Purkinje fibers spread the impulse along the ventricles: ventricular excitation is completed



Electrocardiogram (ECG or EKG)

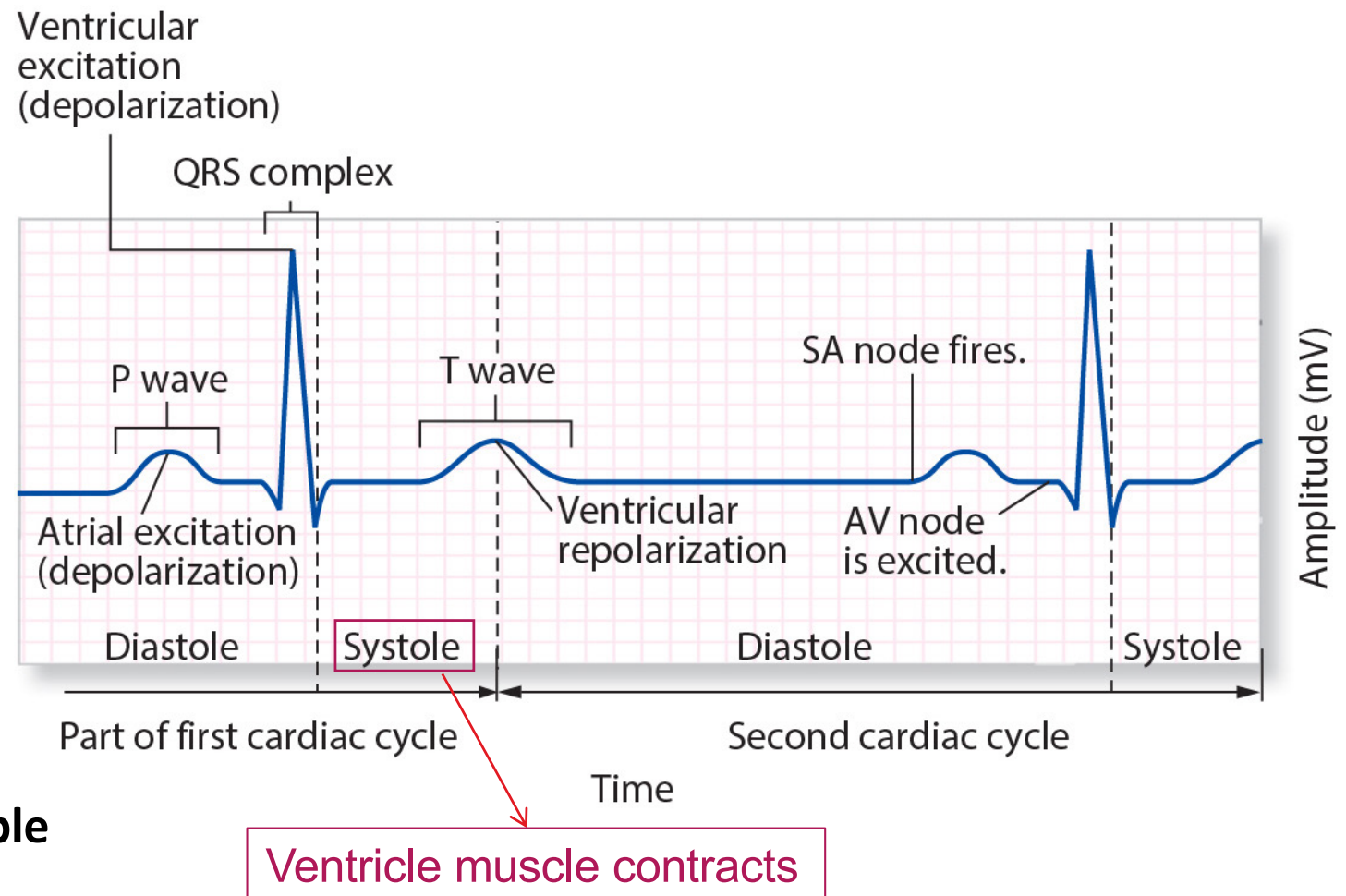
- Record of **electrical impulses generated** during the **cardiac cycle**
- Monitor **electrical activity initiated by SA node**

- **P wave** – begins when SA node generates action potentials → **atrial excitation**

- **QRS complex** – complex wave – AV node **excites ventricles**

- **T wave** – **repolarization of ventricles** back to resting state

- **Repolarization of atria is not detectable (masked)**



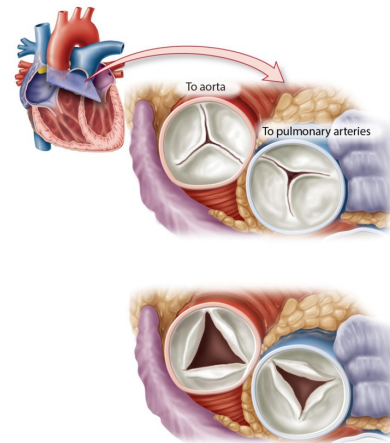
Contraction of the heart

Events that produce a **single heartbeat**

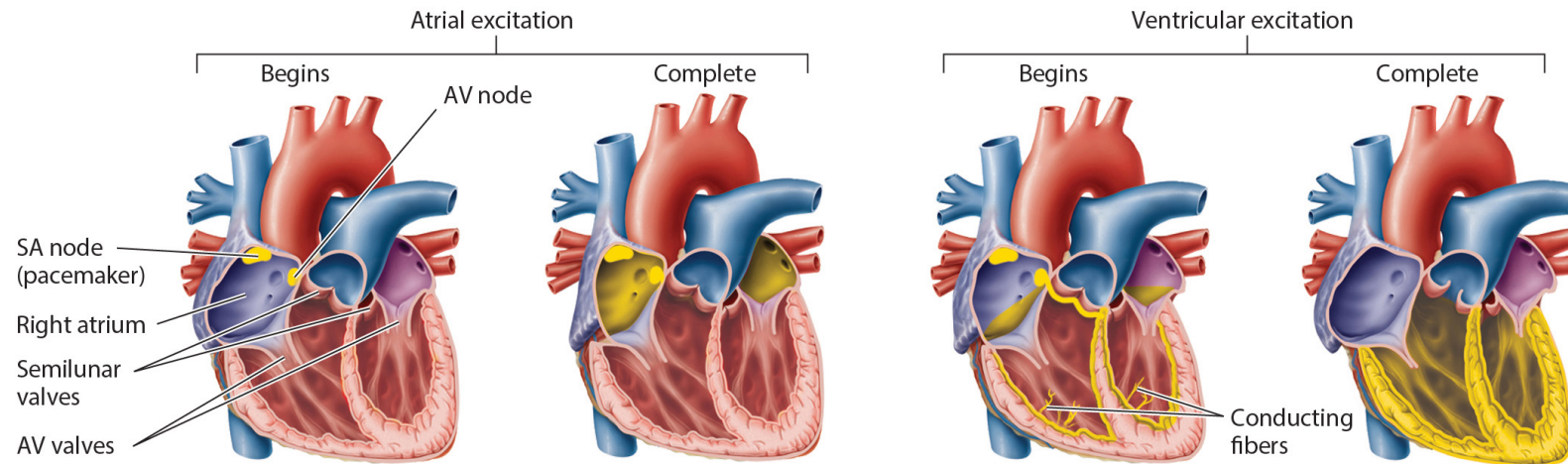
Two phases:

- **(Ventricular) Systole – both ventricles contract** and blood is ejected from the heart (blood **pressure highest**)
- **(Ventricular) Diastole –ventricles fill** and **both atria contract** (blood **pressure lowest**)

One-way heart valves open and shut in response to pressure gradients



Contraction of Heart



AV valves	OPEN	OPEN	Beginning to close	CLOSED
Semilunar valves	CLOSED	CLOSED	CLOSED	OPEN
Phase of cardiac cycle	Diastole	Diastole	Systole beginning	Systole
Atria	Relaxed	Contracted	Beginning to relax	Relaxed
Ventricles	Relaxed	Relaxed	Beginning to contract	Contracted
Chambers with highest pressure	All are low pressure	Atria (slightly)	Ventricles	Ventricles

- **Both atria contract together forcing blood through AV valves into ventricles**
- **Electrical impulses reach atrioventricular (AV) node conducts impulse to ventricles**
- **Both ventricles contract together forcing blood through semilunar valves into systemic or pulmonary arteries**
- **AV valves shut so blood travels only one way**

Electrocardiogram (ECG or EKG)



Blood Vessels

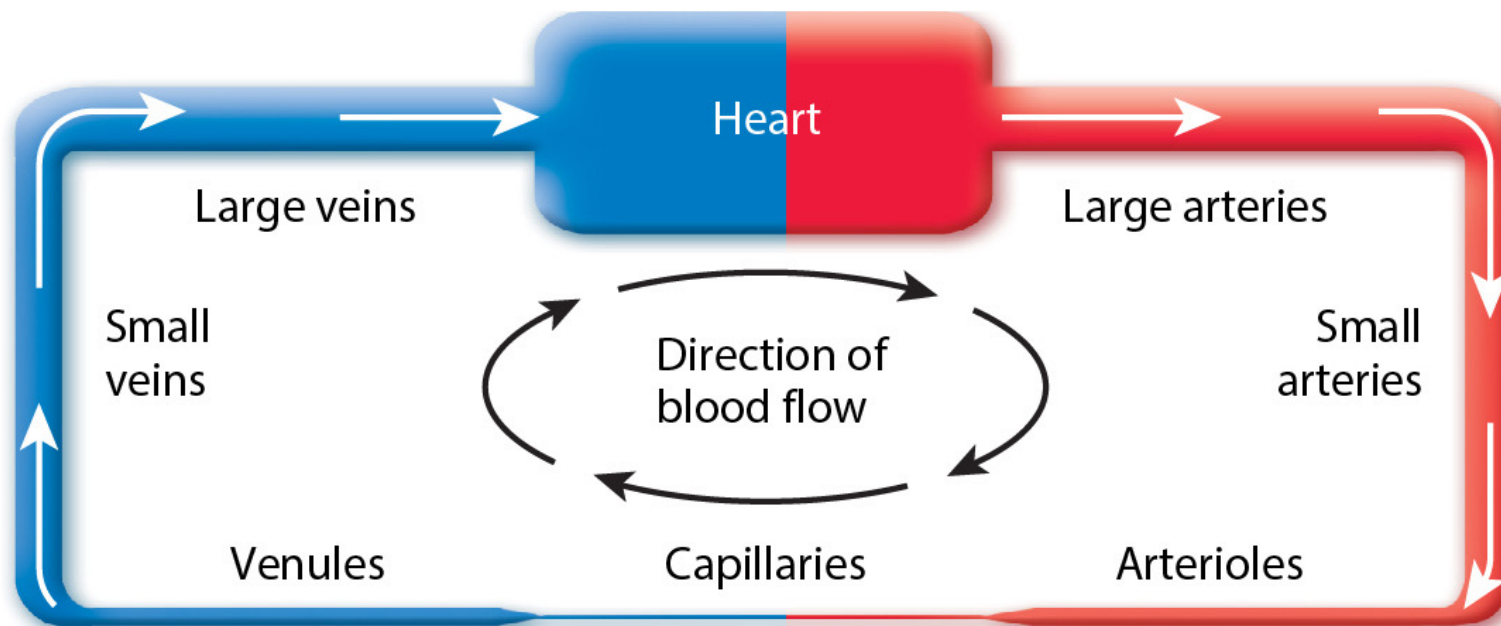
Arteries – Conduct blood away from the heart

Arterioles – Constrict to control blood distribution

Capillaries – Gas, nutrient, and waste exchange

Venules – Small, thin extensions of capillaries

Veins – Conduct blood back to the heart



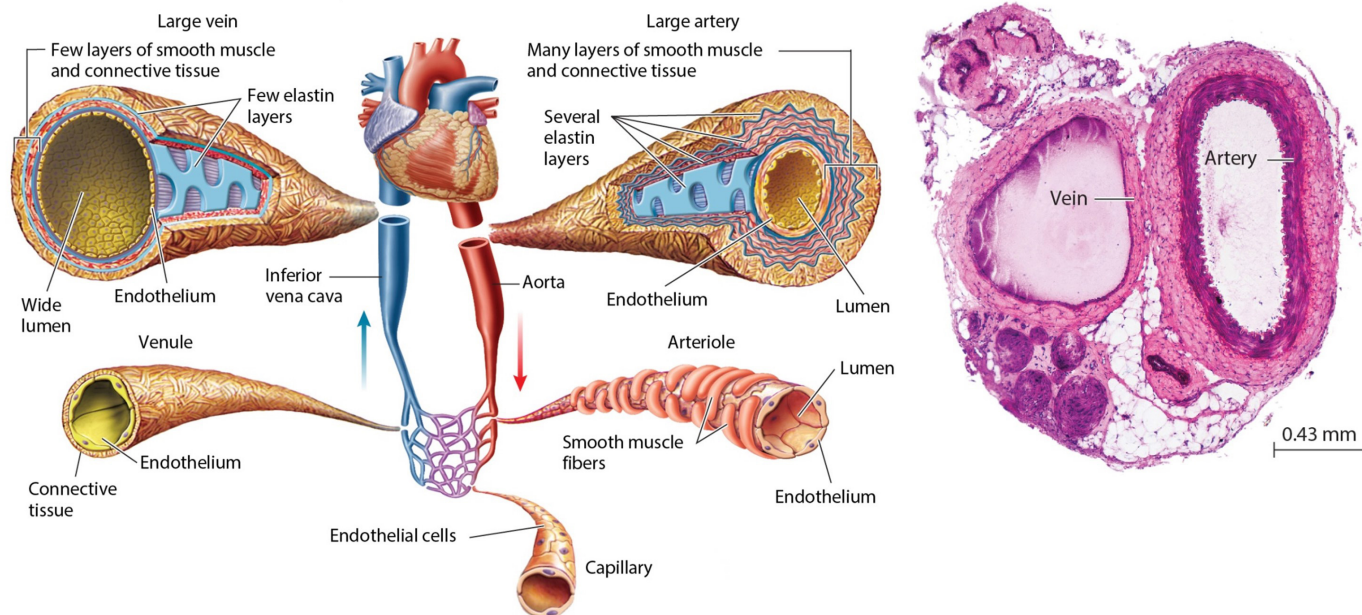
Arteries and Arterioles

Arteries

- Conduct **blood away from the heart**
- **Layers of smooth muscle** and connective tissue **around endothelium**

Arterioles

- Can **dilate or constrict** to **control blood distribution to tissues**



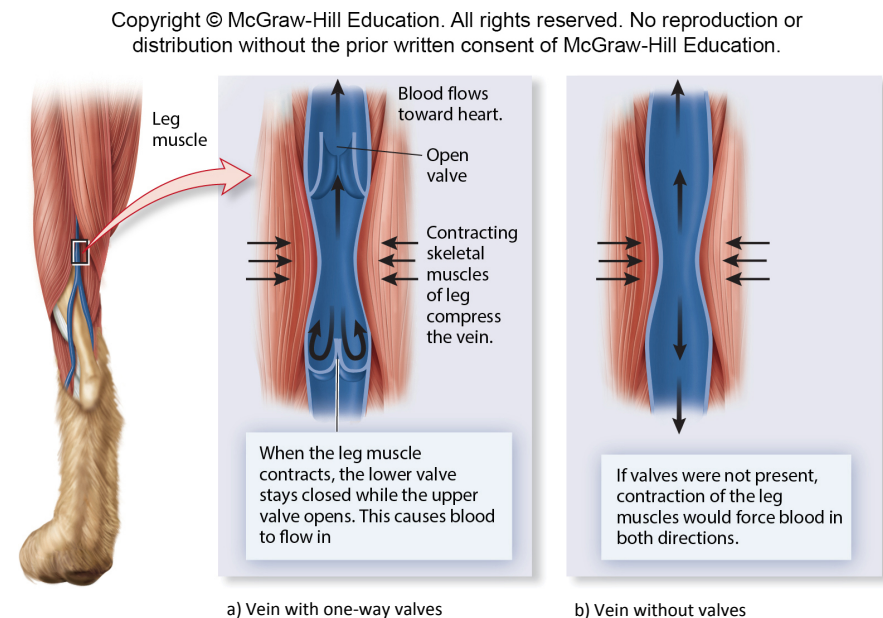
Venules and Veins

Venules

- **Small, thin extensions of capillaries**

Veins

- Conduct blood **back to the heart**
- Thinner and **less muscular than arteries**
- **Need help returning blood to the heart**
 - **Smooth muscle contractions help propel blood**
 - **Valves inside veins** squeezed by skeletal muscles **ensure direction**



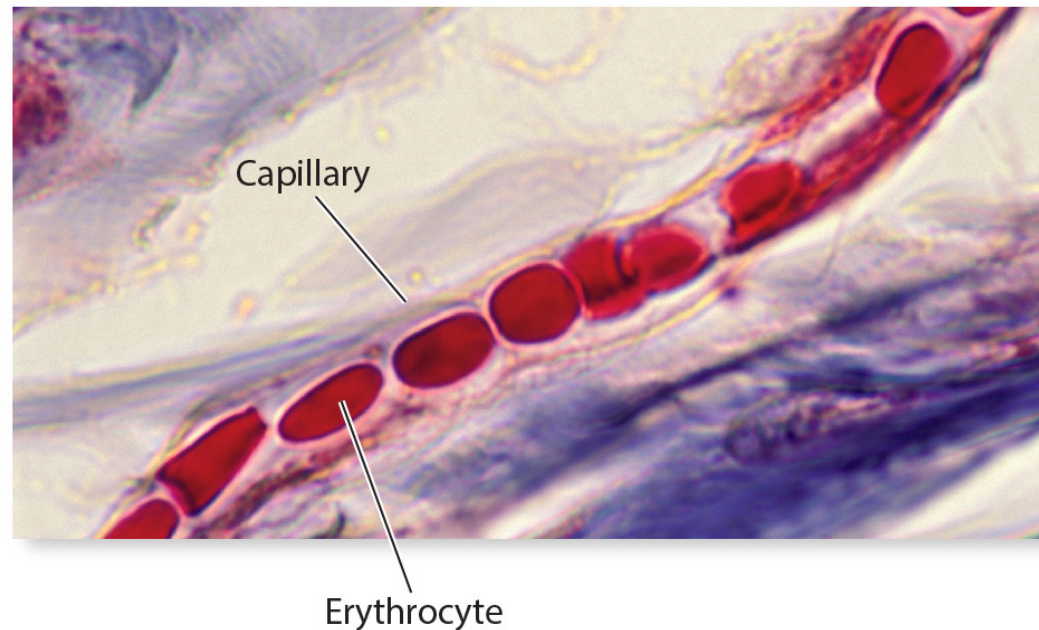
Capillaries

Site of **gas and nutrient/waste exchange**

Single-celled layer of endothelium on a basement membrane

Smallest and narrowest vessels in the body

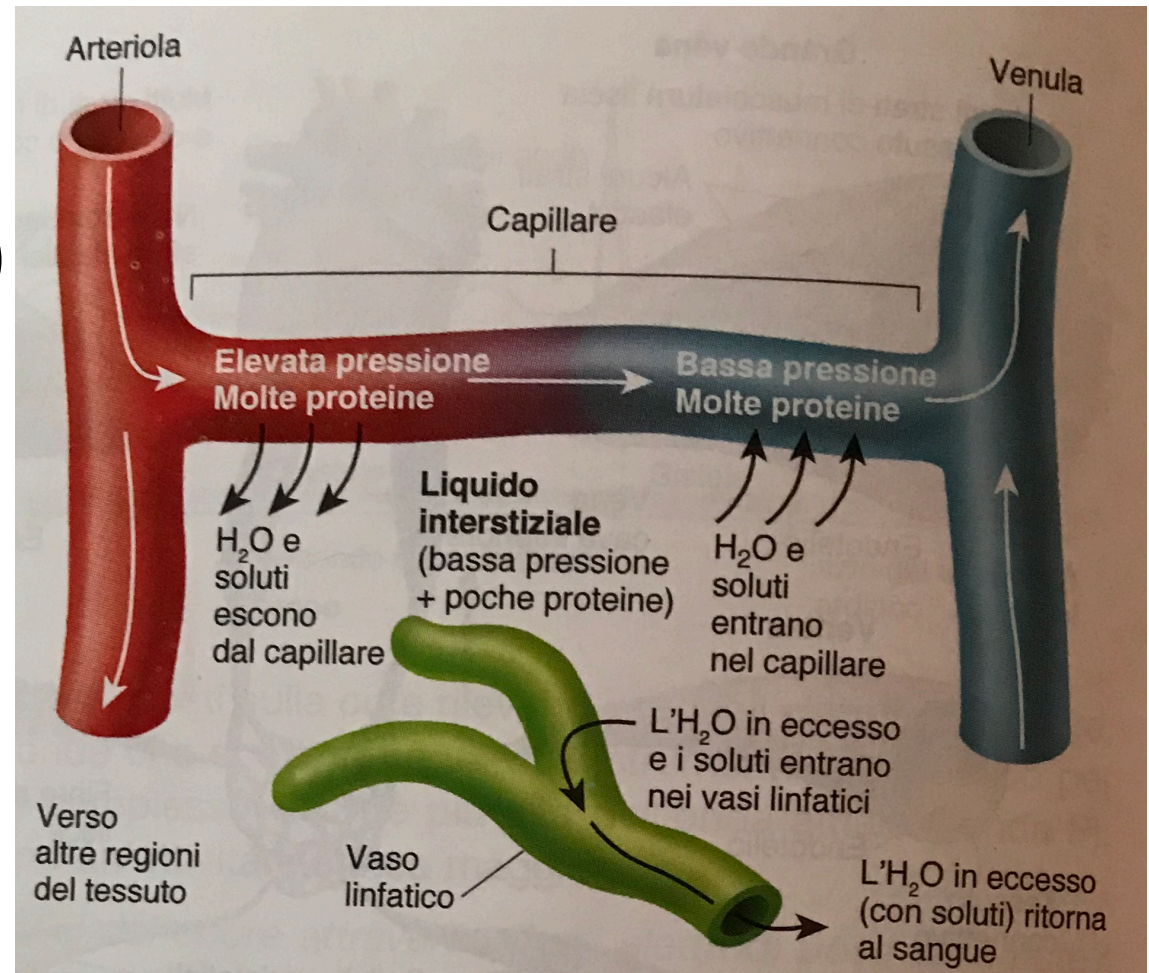
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Pressure Changes in Capillary

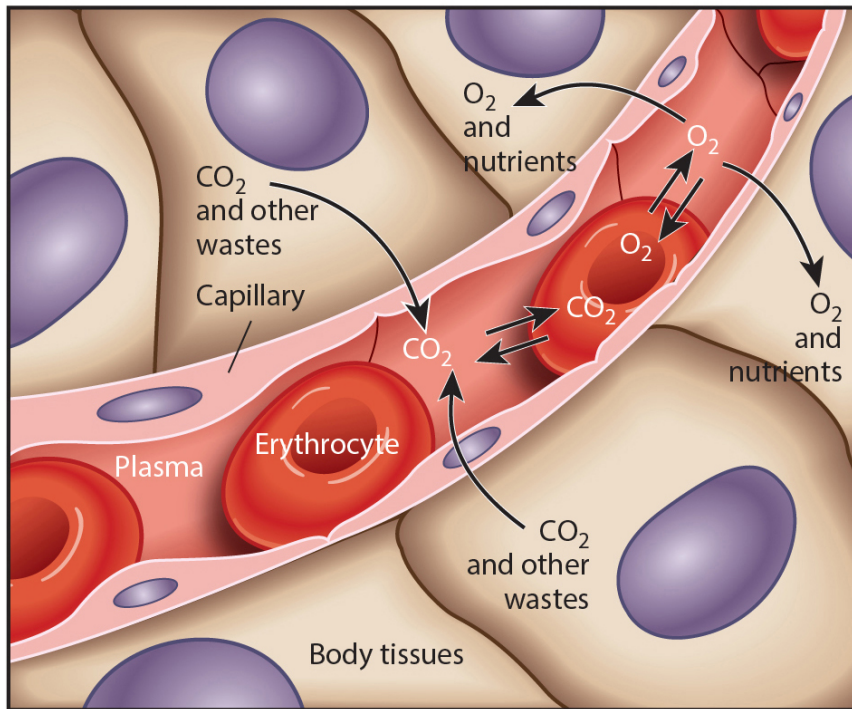
Blood enters capillary under pressure

- **Pressure forces some fluid out of the blood (but not red blood cells or large proteins)**
- **Most of the fluid that leaves is recaptured at the venule end of the capillary**
- **Lymphatic system will collect fluid that is not captured and return it to the blood**



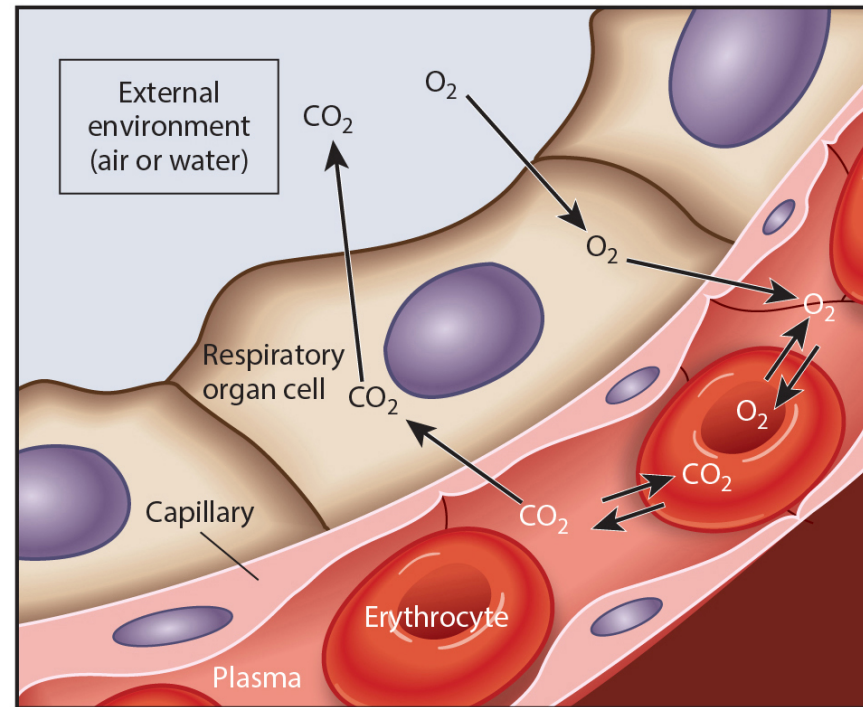
Gas and nutrient/waste exchange in capillaries

Peripheral tissues



b) Gas, nutrient, and waste exchange between the blood and tissues

Respiratory organ (lungs)



a) Gas exchange between the environment and a respiratory organ

Gas and nutrient/waste exchange in capillaries



Mechanisms of Gas Transport in Blood

O₂

- **Not enough oxygen** would **dissolve** directly into **blood** to support metabolic needs
- Respiratory **pigments (hemoglobin)** increase the amount of gas carried

CO₂

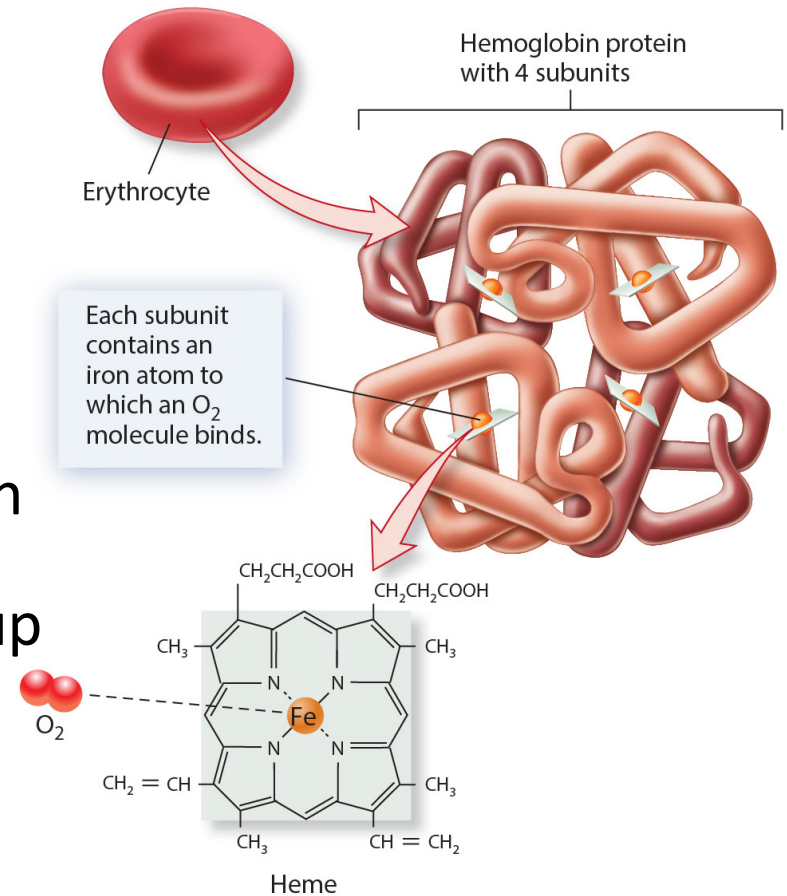
- Only a **limited amount of CO₂** dissolves in blood
- **Some** carried by **hemoglobin**
- **Most** highly **soluble bicarbonate ions** (HCO_3^-)

Respiratory Pigments: hemoglobin

In erythrocytes

Hemoglobin – iron Fe^{2+}

- **4 protein subunits**
- **Each** has a **heme unit** – contains iron
- Single hemoglobin molecule **binds** up to **4 oxygen molecules**



All have a high affinity for oxygen

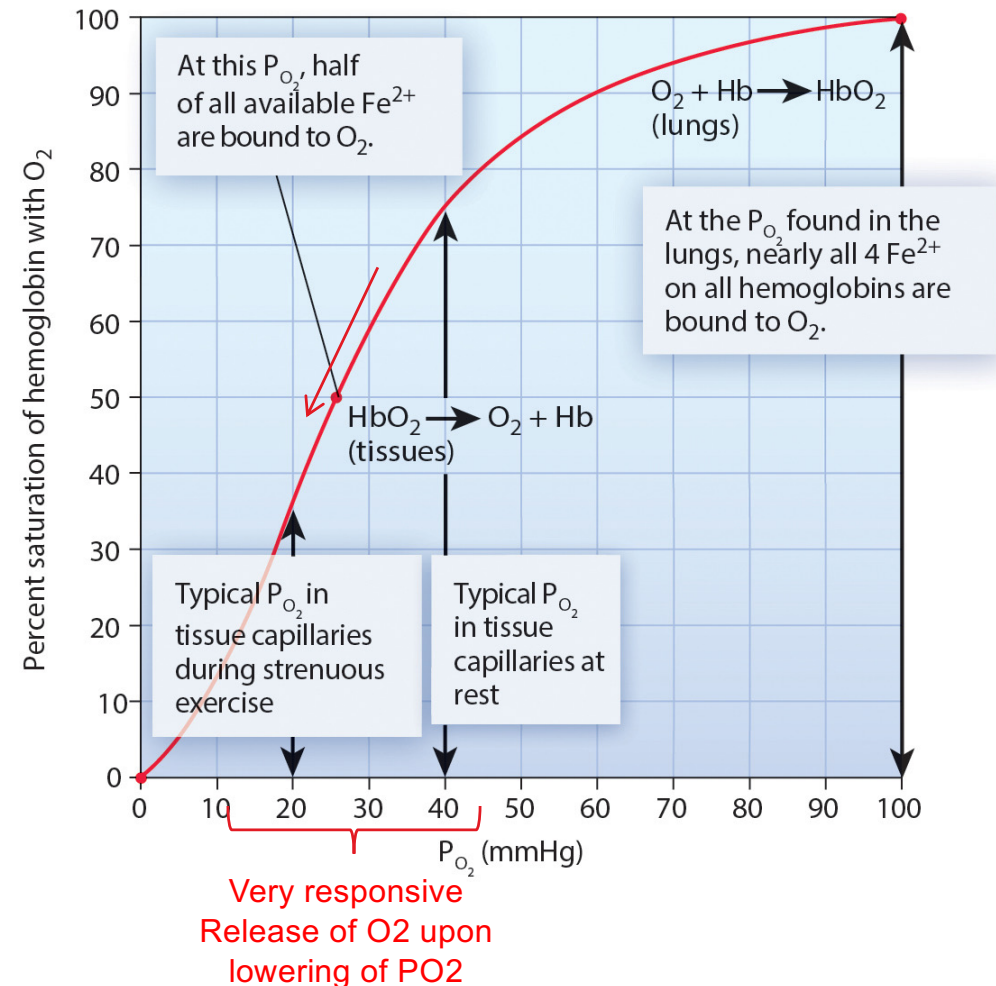
Binding of oxygen is **noncovalent and reversible**

Oxygen-Hemoglobin Dissociation Curve

When partial pressure P_{O_2} is high $\rightarrow$ more O_2 binding

When P_{O_2} is low $\rightarrow$ less O_2 binds to hemoglobin

Sigmoidal curve due to cooperation – shape of hemoglobin changes as oxygen unloads and loads $\rightarrow$ release is **very responsive to changes in PO_2 in low ranges** (non-pulmonary **peripheral tissues**)



Blood Flow, Blood Pressure, and Resistance

Blood pressure (P): force exerted by blood on the walls of blood vessels

Resistance (R): tendency of blood vessels to slow down the flow of blood

- Based on **vessel radius, length and blood viscosity**

Blood flow (F): the movement of blood through the vessels

- $\text{Flow (F)} = \Delta\text{Pressure (P)} / \text{Resistance (R)}$

Control of Blood Flow

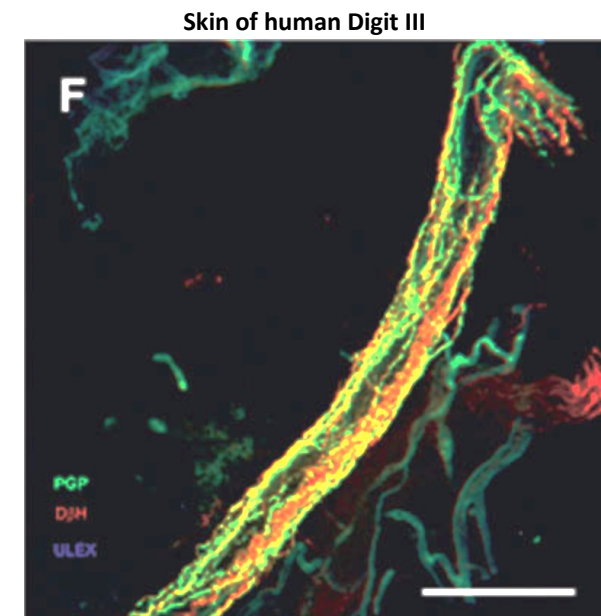
Change in arteriolar resistance is a major mechanism to control blood flow to a region

Radius is the most important factor (inverse to resistance):

- **Vasodilation – increase in radius**
- **Vasoconstriction – decrease in radius**

Controlled by:

- **locally-produced substances**
- **hormones**
- **nervous system input (SNS strongly promotes vasoconstriction)**



Dopamine-hydroxylase (DH): sympathetic neurons
PGP: general neuronal marker
Ulex: binds to the endothelium

Cardiac Output (CO)

Stroke volume = amount of **blood** a heart ejects /beat

Cardiac output = amount of **blood** the heart pumps in liters / minute

- Depends on size of the heart and frequency how it beats

To meet their **high metabolic demands**, small animals often have **high heart rates** → give them a greater cardiac output than would be predicted based on size

Table 48.1 Comparative Features of Representative Mammalian Hearts*

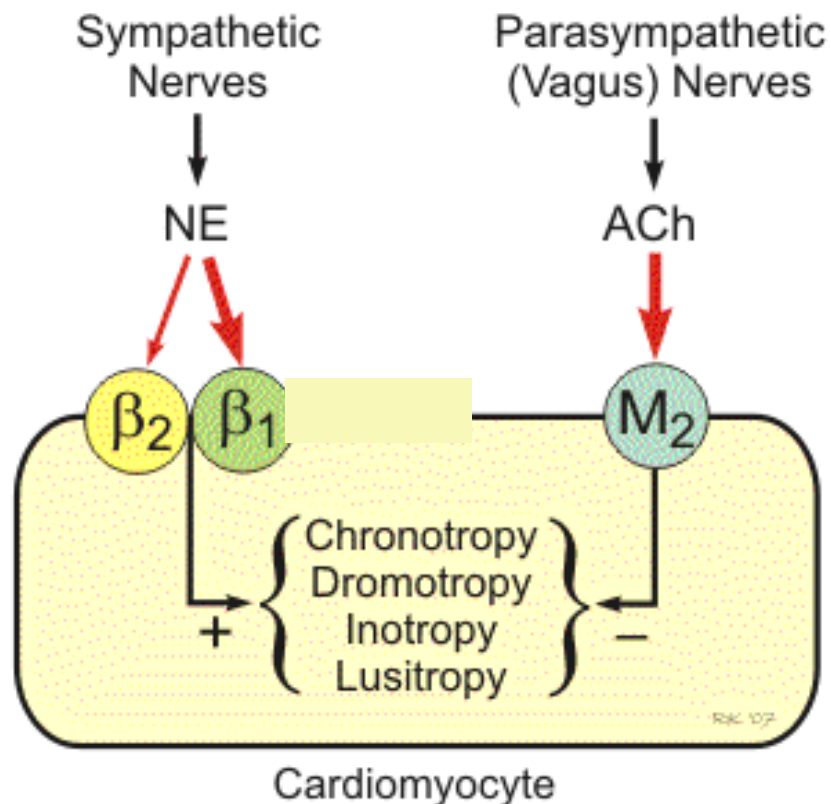
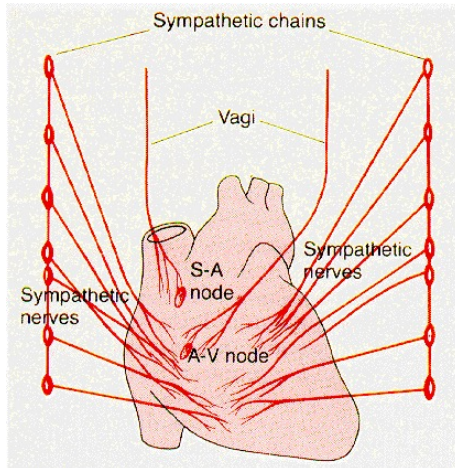
Animal	Body mass (kilogram)	Heart mass (kilogram)	Stroke volume (Liter)	Heart rate (beats per minute) [†]
Shrew ^{††}	0.0024 (2.4 grams)	0.000035 (35 grams)	0.000008 (8 microliter)	835
Rat	0.20	0.001	0.0002	360
Rabbit	2	0.012	0.0013	189
Small dog	5	0.030	0.007	120
Large dog	30	0.180	0.040	88
Human	75	0.380	0.075	70
Horse	450	3.50	0.90	38
Elephant	4,000	25	4.0	25
Blue whale	100,000	600	100	10

*Values are based on average body masses and resting conditions. In some cases, stroke volumes are estimates based on heart size.

[†]bpm = beats per minute.

^{††}The shrew reported here is the Etruscan shrew, one of the smallest known mammals. Its heart is somewhat larger than would be predicted for its body mass. Note its heart rate; at 835 beats per minute, the heart beats 14 times per second!

Autonomic effects on myocardial function



Sympathetic stimulation

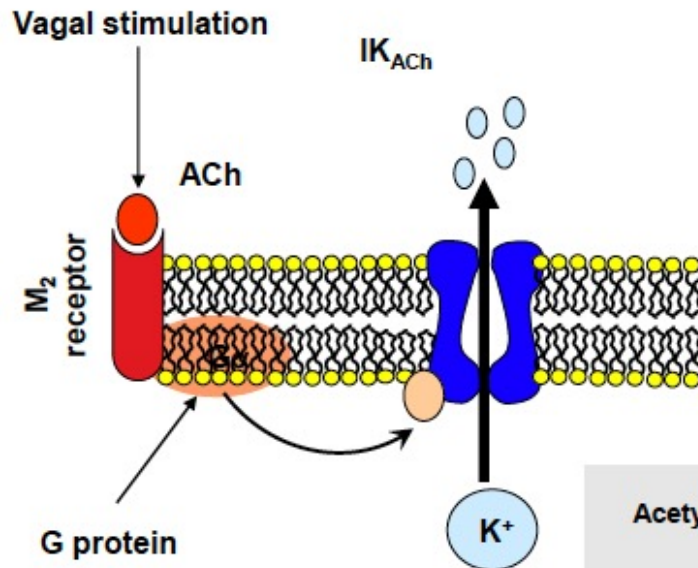
- **Increase heart rate** (Chronotropy)
- **Increase Atrioventricular (AV) conduction** (dromotropy)
- Decreases action potential duration (both in atria and ventricular myocytes) → Generally is **proarrhythmic in both chambers**
- **Increases myocardial contractility** from which **stroke volume** is dependent (inotropy)

Parasympathetic stimulation:

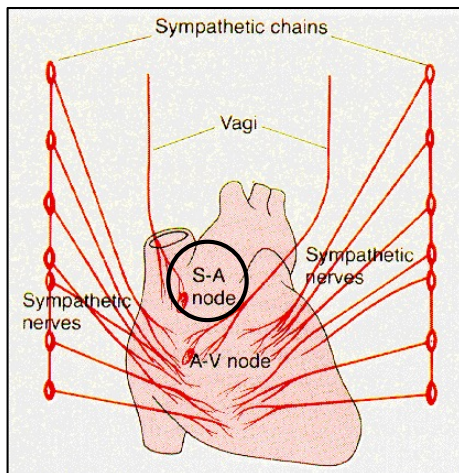
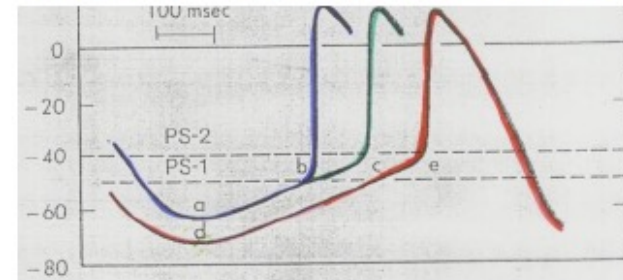
- **Decrease heart rate**
- **Decrease Atrioventricular (AV) conduction**
- Prolongs action potential duration and refractory period in ventricles, but reduces refractory period in atria → **anti-arrhythmic only in ventricles**

Autonomic effects on myocardial function

Ionic basis of sinus bradycardia induced by vagal stimulation



Vagal stimulation:
hyperpolarization of the membrane potential



Acetylcholine release from the nerve endings of the vagal system

↓
acetylcholine acts on muscarinic receptors

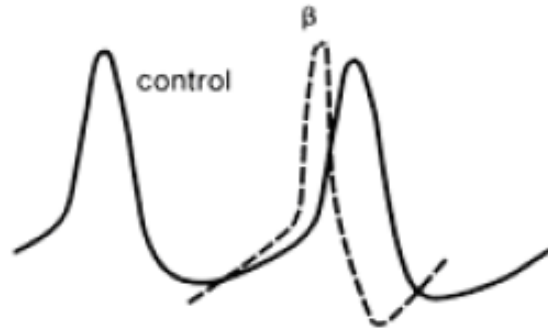
↓
G-protein stimulation

↓
 $I_{K_{ACh}}$ activation (potassium channels activated by acetylcholine)

↓
hyperpolarization of the membrane potential

Autonomic effects on myocardial function

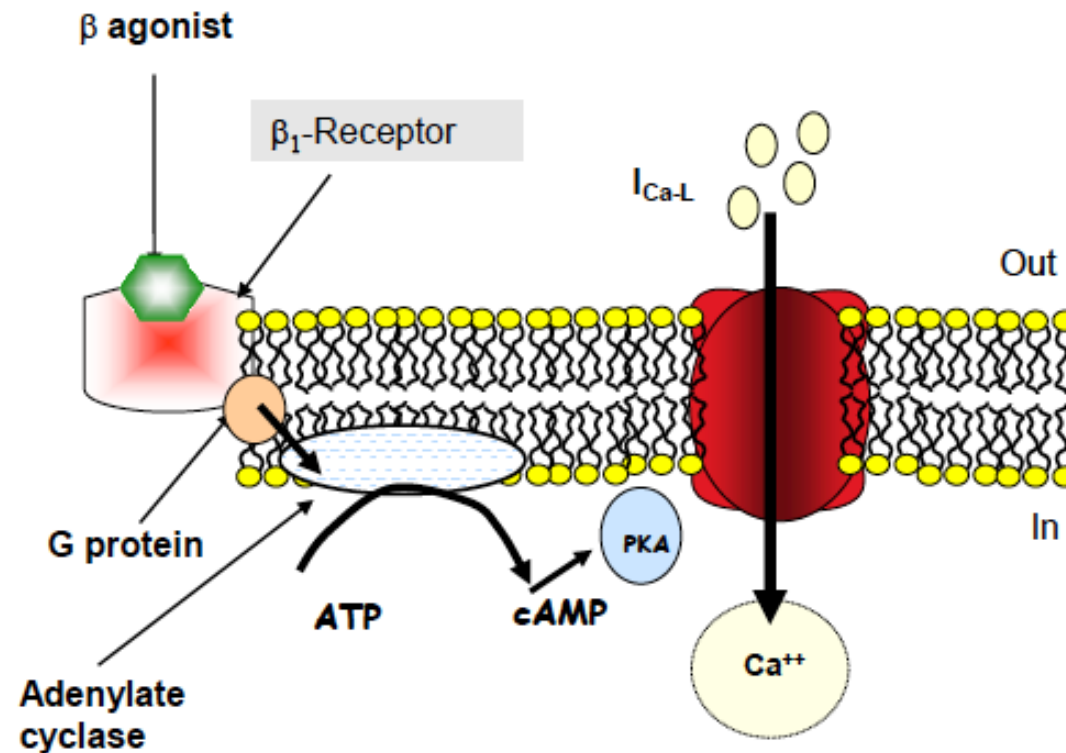
Ionic basis of sinus tachycardia induced by sympathetic stimulation



β -adrenergic stimulation increases the probability of calcium L-channel opening



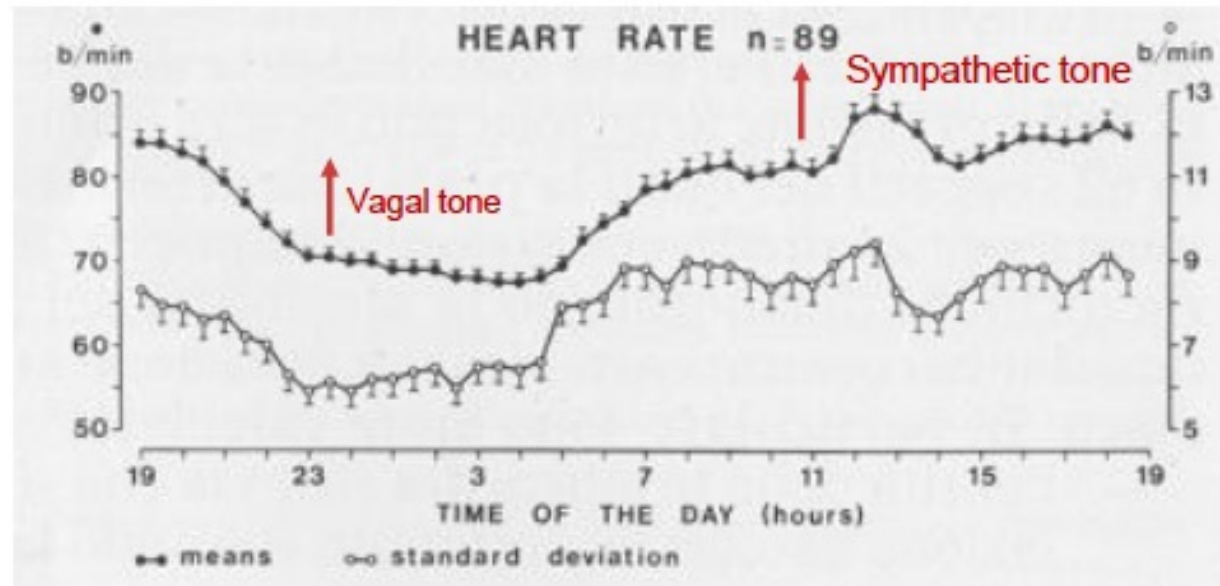
the slope of the diastolic depolarization increases



Autonomic effects on myocardial function

Autonomic circadian modulation of the heart rate

24-hour Holter recording

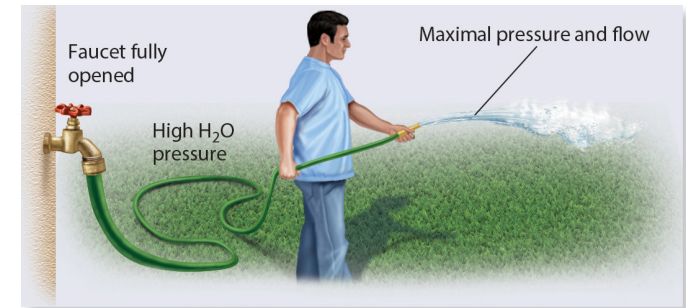


Systemic Blood Pressure

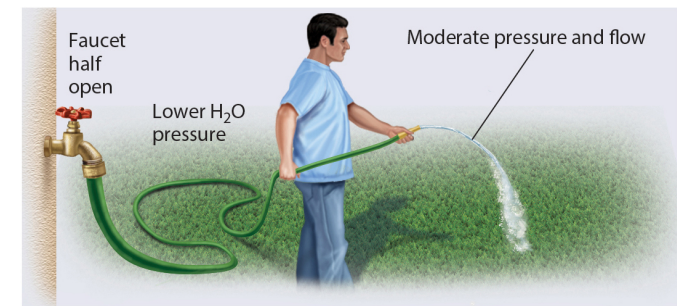
The **greater the cardiac output** and the **higher the resistance**, the **higher the blood pressure**

Garden hose analogy

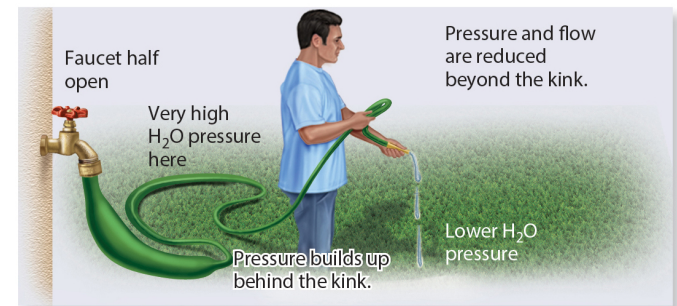
High SNS tone increase both cardiac output and vascular resistance → increase in blood pressure



a) Maximal cardiac output with low resistance

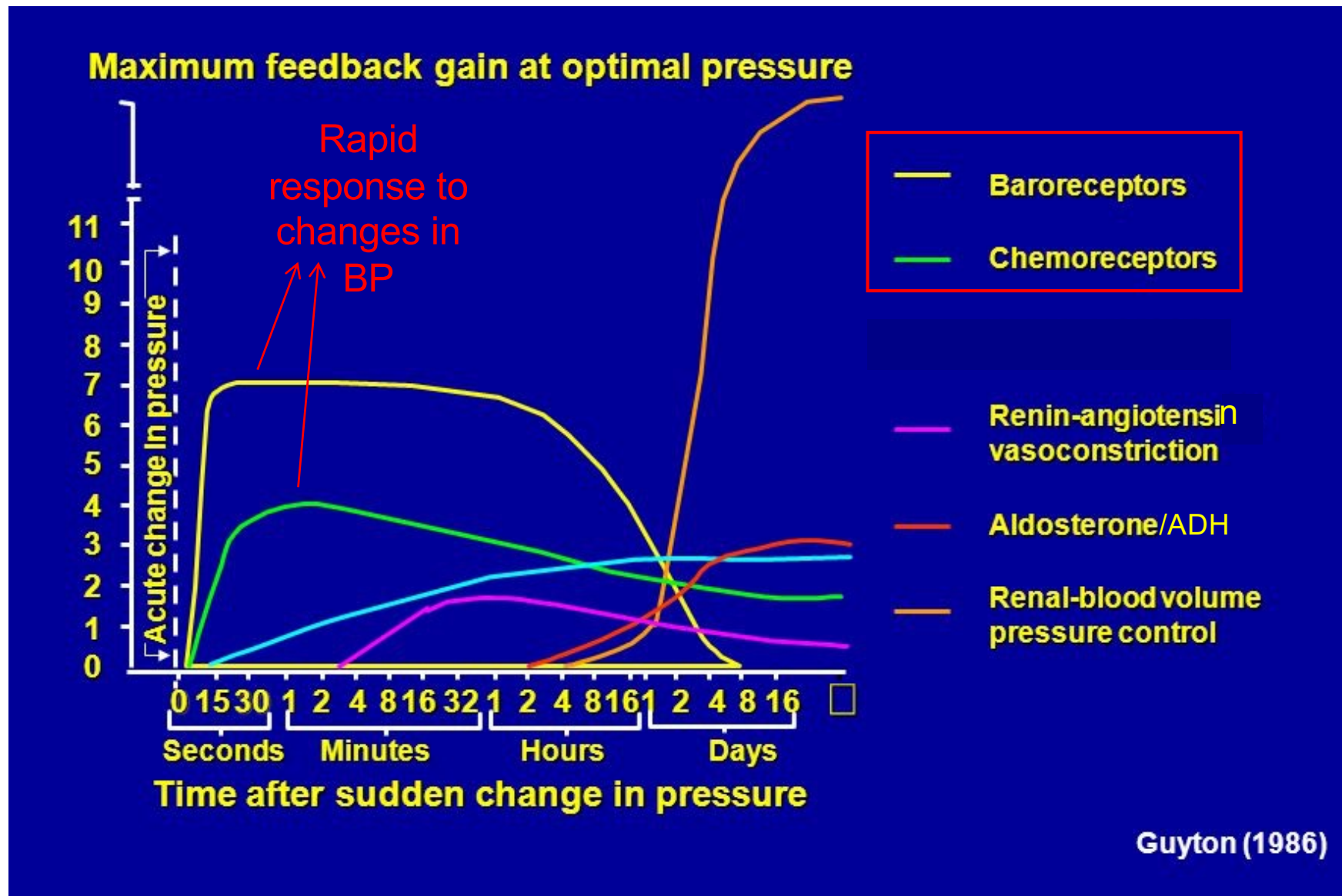


b) Moderate cardiac output with low resistance



c) Moderate cardiac output with high resistance

Mechanisms to control blood pressure



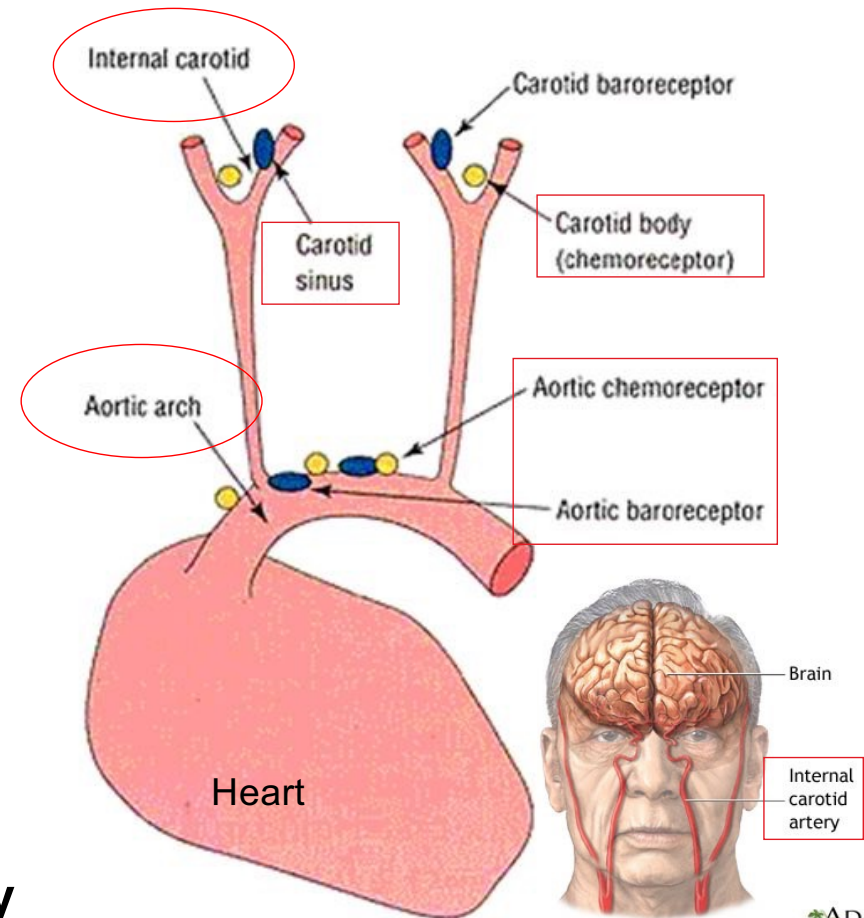
Baro- and chemo-receptors

Baroreceptors

- Detect changes in **blood pressure**
- Located in:
 - **Internal Carotids (carotid sinus)**
 - **Aortic Arches**

Chemoreceptors

- Detect changes in blood **Oxygen, pH/CO₂**
 - Located in:
 - **Internal Carotids (carotid body)**
 - **Aortic Arches**
 - **Central**
- Important in the **control of Respiratory function**

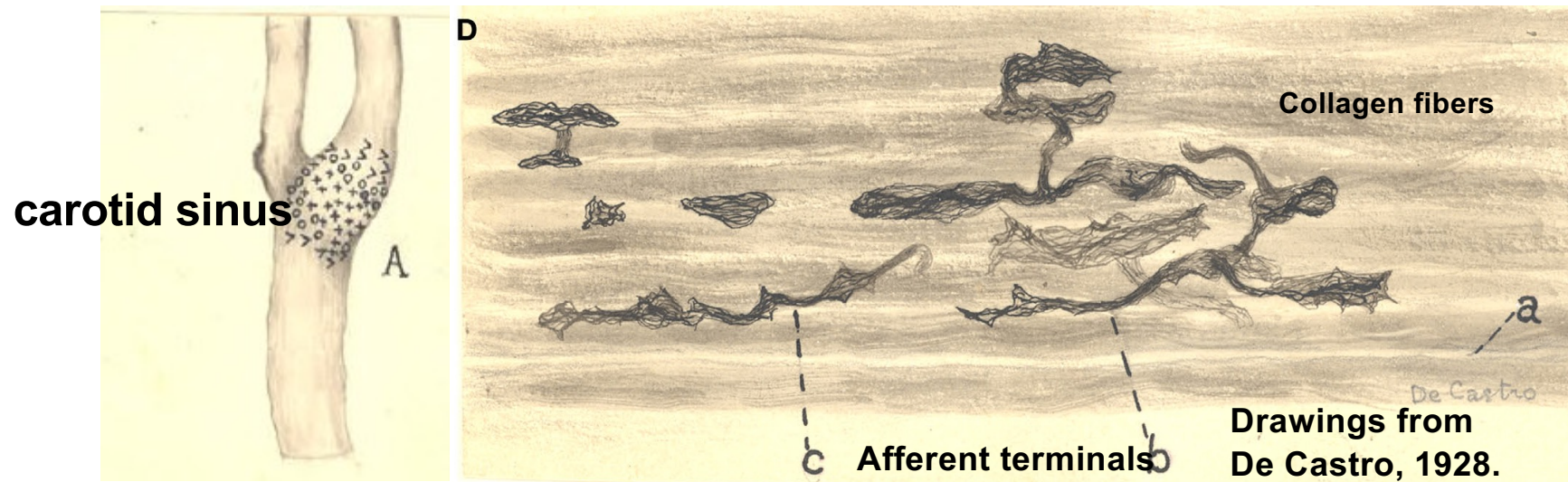


Baroreceptors

Baroreceptor of

- **Internal Carotids (carotid sinus)** → monitor blood pressure entering the brain
- **Aortic Arches** → monitor pressure of blood leaving the heart

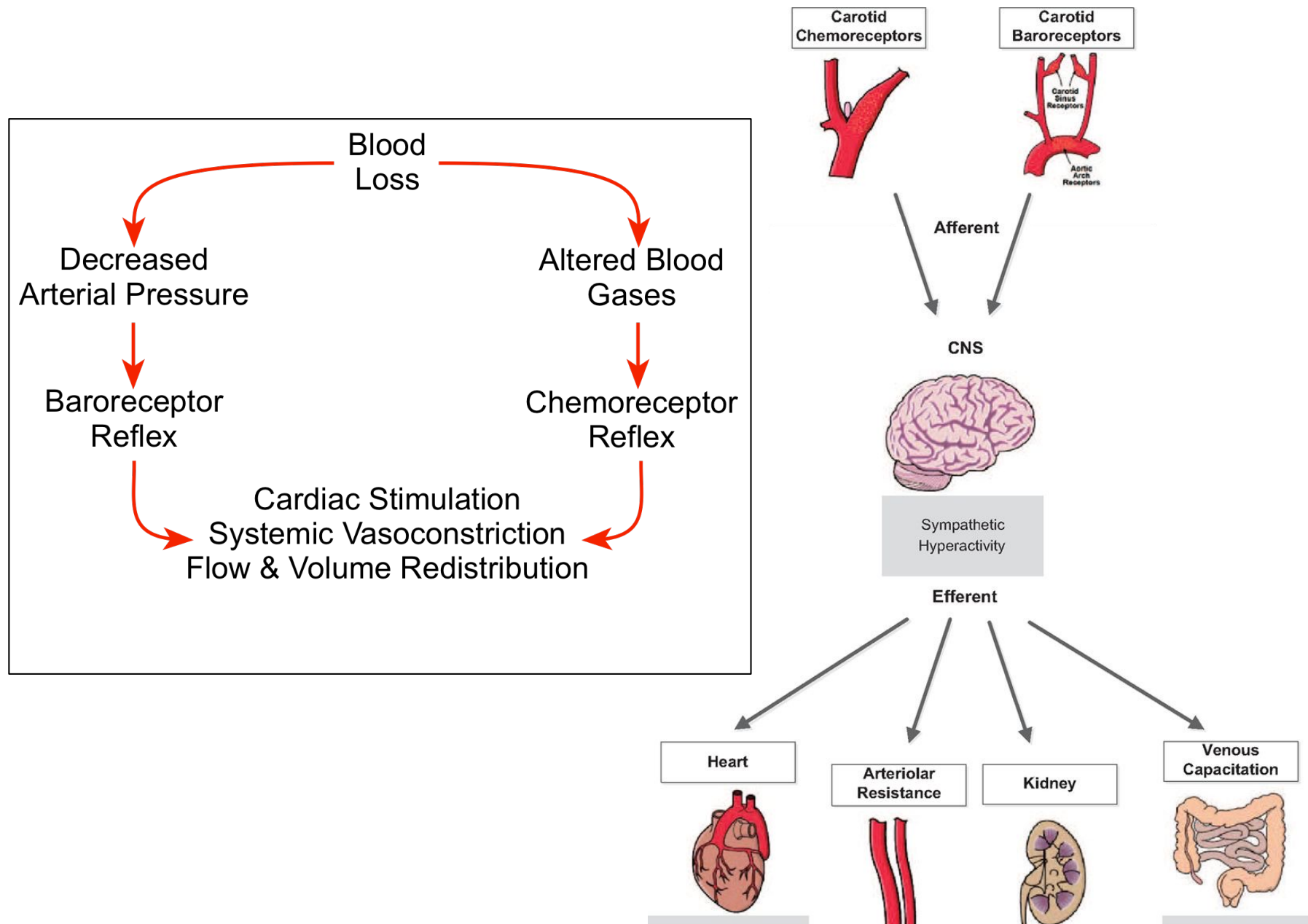
Baroreceptors are afferent **mechanosensitive fibers** interspersed in the **arterial elastic layers** (aorta or carotid sinus)



Baroreceptors may respond to two parameters:

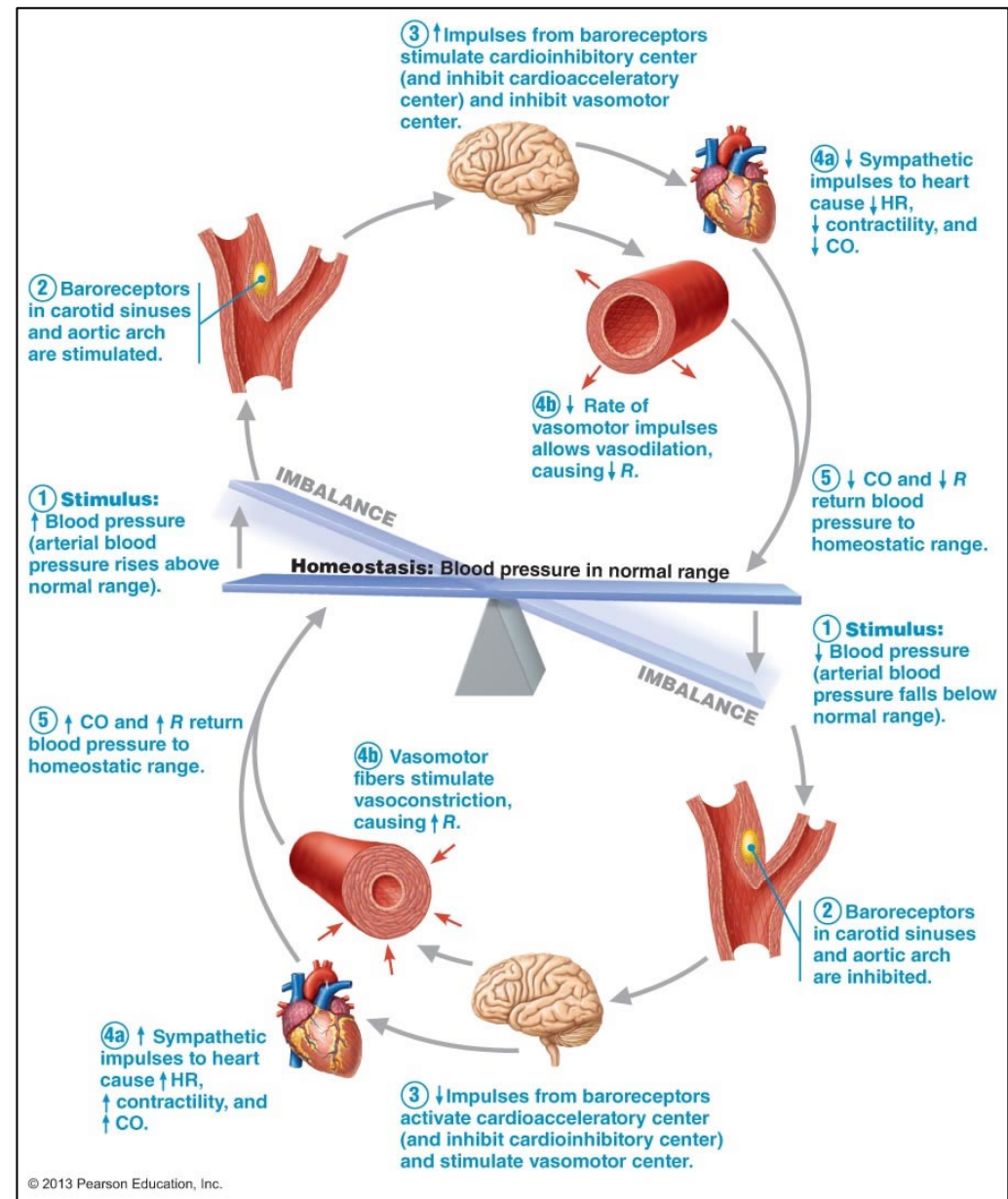
- **absolute stretch** (absolute pressure change)
- **Rate at which stretch is occurring** (rate of pressure change)

Baro- and Chemo-receptor Reflexes control of blood pressure



The baroreceptor reflex and blood pressure homeostasis

- **Increased MAP** results in **decreased sympathetic vasomotor tone** and mediates **vasodilation**. Similarly, decreased MAP results in an increase in sympathetic vasomotor tone and vasoconstriction.
- There are similar directional changes with **sympathetic efferent traffic to heart**. **Augmented cardiac sympathetic** nerve activity **increases heart rate and cardiac output**, decreased sympathetic activity decreases it.
- **Increased MAP** results in an **activation of the parasympathetic** traffic to the heart. Acetylcholine released from parasympathetic fibers on the heart has a **powerful inhibitory affect on the sino-atrial node** and therefore **slows the heart rate**.

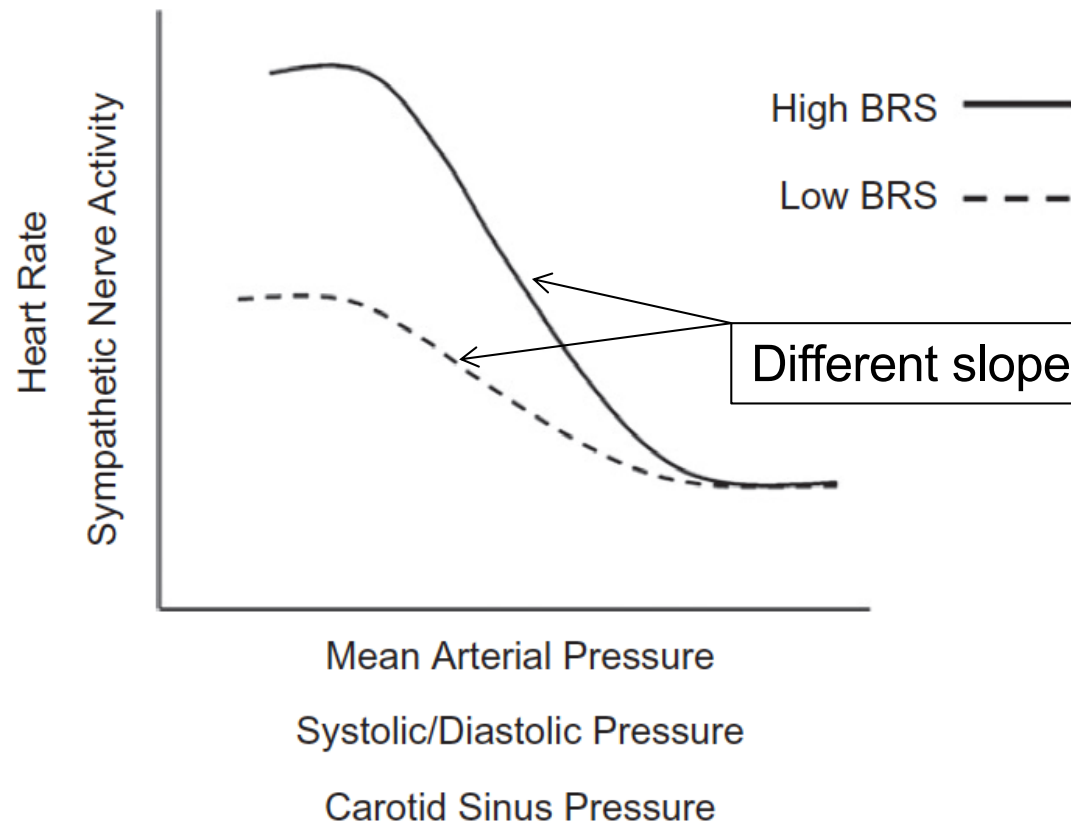


The baroreceptor reflex and blood pressure homeostasis



Baroreflex sensitivity (BRS)

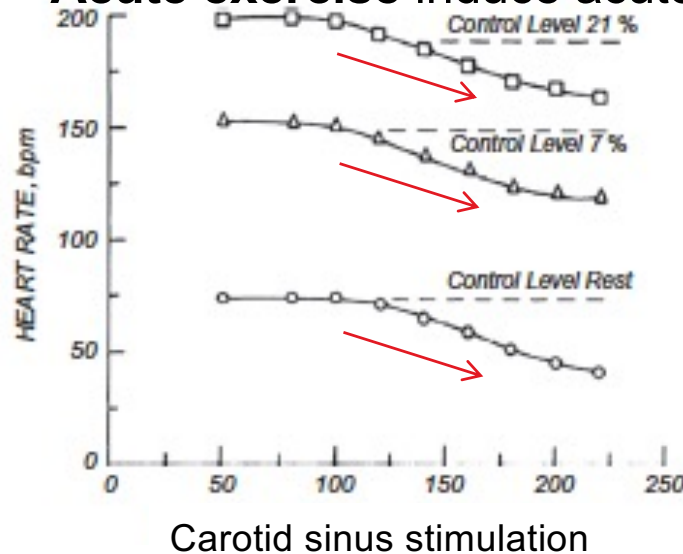
- BaroReflex Sensitivity (BRS) is given by the **slope of the linear portion** of the sinusoidal response of Heart Rate or Sympathetic Nerve activity to an increase in blood pressure.



- Changes in sensitivity** are associated with pathologies with **poor pressure homeostasis**

Baroreflex resetting (acute) during exercise

- During acute physical exercise there is an increase in blood pressure
- **Acute exercise** induce acute resetting:



Stimulation of carotid sinus induce similar heart rate changes in dogs during rest or mild (7% slope on treadmill) or moderate (23% slope) exercise. Note the **shift of the curves, without changes in the curve characteristics** indicating **unaltered baroreflex sensitivity** (Melcher 1981)

- The resetting allow the baroreflex to **work over a dynamic range (away from curve plateau)**
- The shift allows the **higher pressure to be maintained more easily**
- May be dependent on changes in afferent (sensory), central and efferent (autonomic) activity.
- Acute resetting is **reversible**

Impact on Public Health

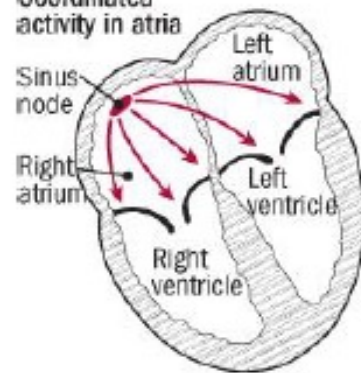
Cardiovascular disease (conditions affecting the heart and blood vessels) account for **more deaths each year in the US than any other cause**

Arrhythmias: Spiral-Wave Reentry and Electrical Turbulence

Sinus rhythm

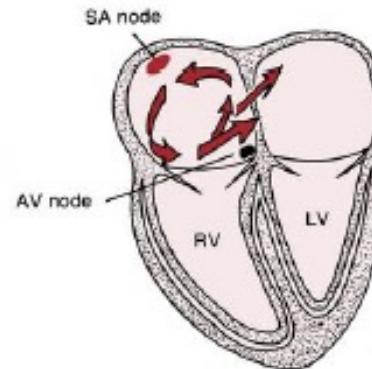
A. Normal sinus rhythm

Coordinated activity in atria



Tachycardia: spiral wave reentry

ATRIAL FLUTTER



NSR

VT

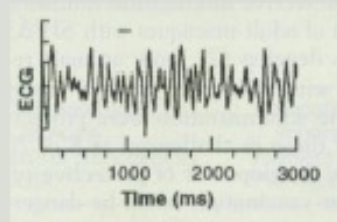
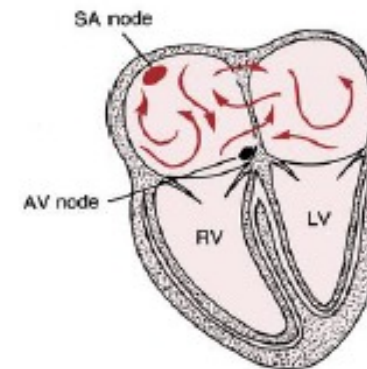
VF

(a)



Fibrillation: multiple wavelets (turbulence)

ATRIAL FIBRILLATION



Experimental evidence of spiral waves in the isolated rabbit heart

Gray RA, Jalife J, Panfilov AV et al. Science, 1995

Hypertension and Atherosclerosis

Systemic hypertension (high blood pressure) causes damage leading to the formation of plaque inside artery walls, blocking the lumen

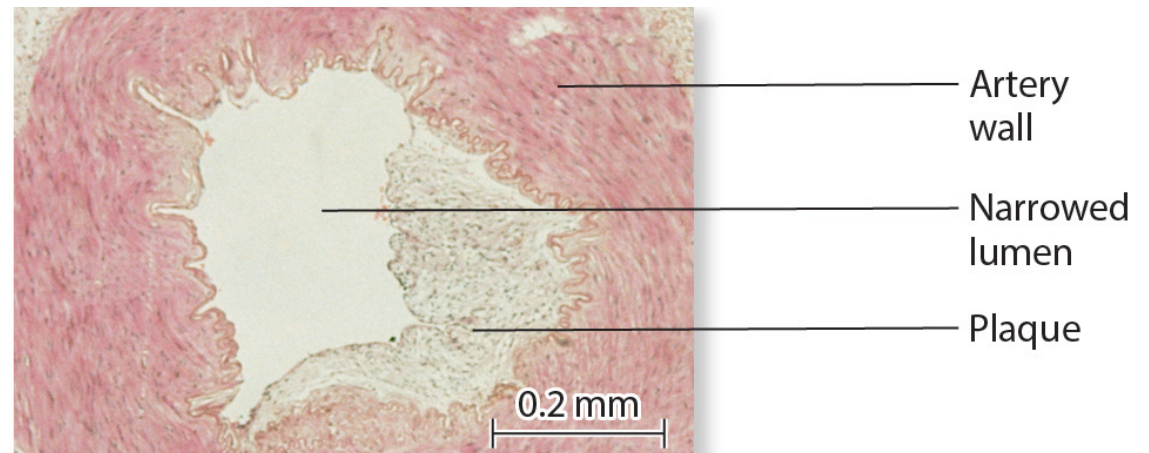
Rarely has any noticeable symptoms

- Often referred to as a “**silent killer**”

Plaques may lead to atherosclerosis

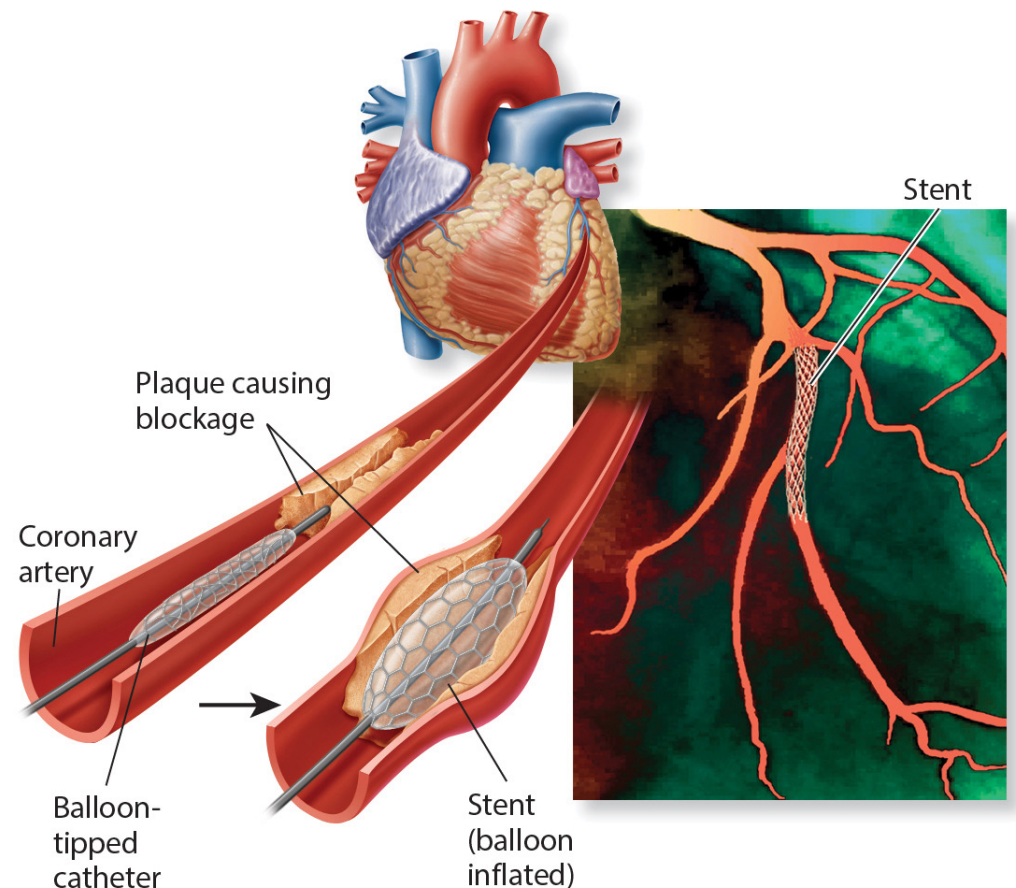
- Plaque causes the artery to narrow and harden
- Large plaques may occlude (block) the lumen of an artery entirely

Coronary artery disease results from **plaque forming in the coronary arteries** supplying the heart muscle



Myocardial Infarction (MI)

- Localized **regions of heart muscle die** when the **blood supply (coronary arteries)** is cut off
- Dead cardiac muscle does not regenerate
- **Cardiac angiography** can detect narrowing of coronary vessels
- **Balloon angioplasty** can widen the lumen of narrowed vessels
- **Coronary artery bypass** uses a healthy blood vessel to replace a blocked coronary artery



Chapter 48

Lecture Outline

See separate PowerPoint slides for all figures and tables pre-inserted into PowerPoint without notes.

